

Cortico-striatal action control inherent of opponent cognitive-motivational styles

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Abstract

Turning on cue or stopping at a red light requires the detection of such cues to select action sequences, or suppress action, in accordance with cue-associated action rules. Cortico-striatal projections are an essential part of the brain's attention-motor interface. Glutamate-sensing microelectrode arrays were used to measure glutamate transients in the dorsomedial striatum (DMS) of male and female rats walking a treadmill and executing cued turns and stops. Prelimbic-DMS projections were chemogenetically inhibited to determine their behavioral necessity and the cortico-striatal origin of cue-evoked glutamate transients. Furthermore, we investigated rats exhibiting preferably goal-directed (goal trackers, GTs) versus cue-driven attention (sign trackers, STs), to determine the impact of such cognitive-motivational biases on cortico-striatal control. GTs executed more cued turns and initiated such turns more slowly than STs. During turns, but not missed turns or cued stops, cue-evoked glutamate concentrations were higher in GTs than in STs. In conjunction with turn cue-evoked glutamate spike levels, the presence of a single spike rendered GTs to be almost twice as likely to turn than STs. In contrast, multiple glutamate spikes predicted GTs to be less likely to turn than STs. In GTs, but not STs, inhibition of prelimbic-DMS projections attenuated turn rates, turn cue-evoked glutamate peaks, and increased the number of spikes. These findings suggest that turn cue-evoked glutamate release in GTs is tightly controlled by cortico-striatal neuronal activity. In contrast, in STs, glutamate release from DMS glutamatergic terminals may be regulated by other striatal circuitry, preferably mediating cued suppression of action and reward tracking.

Significance Statement

Adaptive behavior involves the selection of behaviorally significant cues and the capacity of selected cues to control behavioral action. Neuronal projections from cortex to striatum are essential for such an integration of attentional with motor functions. Here we demonstrated that glutamate release from cortico-striatal projections primarily influences cued turns but not cued suppression of actions (cued stops). Cortico-striatal control of cued turning was especially powerful in rats which, as a psychological trait, preferably deploy goal-directed attention. Together, our findings demonstrate the role of cortico-striatal input in cued action selection, and they emphasize the experimental and biopsychological significance of investigating the brain's attentional-motor interface in the context of broader cognitive-motivational styles.

eLife assessment

This **valuable** manuscript investigated the role of glutamate signaling in the dorsomedial striatum of rats in a treadmill-based task and reported that it differs in goal-trackers compared to sign-trackers in a way that corresponds to differences in behaviour. The evidence supporting these claims is **solid** but could be further strengthened by adding more analyses and more detailed descriptions of current analyses. These findings will primarily be of interest to behavioural neuroscientists.

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Introduction

Cue-triggered selection of action, or suppression of action, are essential components of adaptive behavior. Fronto-striatal projections insert information about attended cues into the striatum to prioritize cue-linked action, and to facilitate modification of action selection in response to changing action outcomes (Balleine and O'Doherty, 2010 [↗](#); van Schouwenburg et al., 2012 [↗](#); Chatham et al., 2014 [↗](#); Hart et al., 2018a [↗](#); Hart et al., 2018b [↗](#)). Deficient or biased cortico-striatal cue import has been postulated to cause neuropsychiatric symptoms ranging from complex movement control deficits in Parkinson's Disease (PD) to compulsive addictive drug use (Volkow et al., 2006 [↗](#); Bohnen et al., 2009 [↗](#); Ersche et al., 2011 [↗](#); Marshall and Ostlund, 2018 [↗](#); Rasooli et al., 2021 [↗](#); Sarter et al., 2021 [↗](#)).

The cortical control of striatal action selection has been extensively supported by studies investigating the effects of manipulation of the excitability of fronto-striatal neurons in non-human animals (for review see Sharpe et al., 2019 [↗](#)) and fronto-cortical functional connectivity in humans (Postuma and Dagher, 2006 [↗](#); Shepherd, 2013 [↗](#); Devignes et al., 2022 [↗](#)). The present experiments were designed to directly determine the fronto-striatal representation of action-initiating and action-suppressing movement cues. Focusing on the prelimbic projections to DMS (Mailly et al., 2013 [↗](#)), that previously were demonstrated to be necessary for learning goal-directed actions (Hart et al., 2018b [↗](#); Choi et al., 2023 [↗](#)), we measured real-time, cue-locked glutamate concentrations in the DMS using amperometry and glutamate-sensing electrodes. Furthermore, we inhibited prelimbic-DMS projections to attribute such release to activity in this pathway.

Glutamate transients were recorded in rats performing visual and auditory cue-triggered turns and stops while walking a treadmill that stopped and restarted in reverse or same direction, respectively. The development of the Cued-Triggered Turning Task (CTTT; Avila et al., 2020 [↗](#)), was inspired by evidence showing that PD fallers exhibited deficient turns (Stack and Ashburn, 2008 [↗](#); Cheng et al., 2014 [↗](#)). In PD patients, the propensity for falls has been attributed to loss of cholinergic neurons innervating cortical areas and the resulting failure to select and evaluate movement cues (Bohnen et al., 2009 [↗](#); Yarnall et al., 2011 [↗](#); Rochester et al., 2012 [↗](#); Bohnen et al., 2019 [↗](#); Kim et al., 2019 [↗](#); Sarter et al., 2021 [↗](#)). Supporting the validity of the CTTT, in terms of revealing clinically relevant, attentional-motor deficiencies, rats modeling PD falls (Kucinski et al., 2013 [↗](#); Sarter et al., 2014 [↗](#); Kucinski et al., 2015 [↗](#)) were demonstrated to exhibit deficits in cued turning, but not cued stopping (Avila et al., 2020 [↗](#)).

Performing well-practiced, visually or auditorily cued turns and stops requires sustaining attention to cue sources, the continued selection of cues as signified by the cue-triggered behavior (termed cue detection; Posner et al., 1980 [↗](#)), monitoring the state of the treadmill and action

outcome, and maintaining cue modality-linked rules in working memory. Cortico-striatal control of action selection was expected to reflect the capacity for such voluntary, or top-down, attentional control.

To reveal the impact of variations in such capacity, we investigated rats that exhibit, as a trait, opponent attentional biases. These rats are selected from outbred populations using a Pavlovian Conditioned Approach (PCA) test. Goal-trackers (GTs) orient to a Pavlovian reward cue, that is, they learn its predictive significance, but they do not approach and contact such a cue. In contrast, sign-trackers (STs) approach and contact such a cue, which has been interpreted as assigning incentive salience to a Pavlovian cue and attributed to cue-evoked mesolimbic dopamine signaling (Flagel et al., 2011 [↗](#); Flagel and Robinson, 2017 [↗](#); Iglesias et al., 2023 [↗](#)). The relatively greater vulnerability of STs for addictive drug taking and relapse in the presence of Pavlovian drug cues has been extensively documented (e.g., Saunders and Robinson, 2010 [↗](#); Yager and Robinson, 2010 [↗](#); Yager and Robinson, 2013 [↗](#)).

Goal- and sign-tracking index the presence of broader, opponent cognitive-motivational styles that are dominated by a bias toward deploying top-down (or goal-directed) and bottom-up (or stimulus-driven) processing of action cues, respectively (for review see Sarter and Phillips, 2018 [↗](#); for evidence of the presence of these traits in humans see Schad et al., 2020 [↗](#); Colaizzi et al., 2023 [↗](#)). As such biases of GTs and STs were previously shown to be mediated in part via contrasting cholinergic capacities for the detection of cues (Paolone et al., 2013 [↗](#); Koshy Cherian et al., 2017; Pitchers et al., 2017b [↗](#); Pitchers et al., 2017a [↗](#)), we hypothesized that contrasts in the cortico-striatal processing of movement cues contribute to the expression of these opponent biases. The present results indicate that, in GTs, cued turning depends on cortico-striatal glutamate signaling. In contrast, glutamate signaling in STs preferably mediates cued stops and is independent of cortico-striatal neuronal activity.

Materials and Methods

Subjects

378 Sprague Dawley rats (215 females; 250-500 g; obtained from Inotiv, West Lafayette, IN, and Taconic, Rensselaer, NY) were individually housed on a 12-hour light/dark cycle (lights on at 7:00 AM) at ~21°C with *ad libitum* access to food (Laboratory Rodent Diet 5001, LabDiet) and water. The experiments detailed below used four separate cohorts of rats (each consisted of 52 rats, 26 females and composed of rats obtained from both Taconic and Inotiv). The four cohorts were obtained and formed across 19 months, reflecting the experimental demands of amperometric recordings. All experimental procedures were approved by the University Committee on the Use and Care of Animals (UCUCA) at the University of Michigan and carried out in laboratories accredited by the Association for Assessment and Accreditation of Laboratory Animal Care (AAALAC). Animals acclimated to housing quarters for two days before the onset of experimental procedures.

Behavioral phenotyping

Apparatus

Prior to the onset of PCA testing, rats were handled daily for 3 days and given ~7 banana-flavored sucrose pellets (45 mg; BioServ). Rats were tested in conditioning chambers (20.5 x 24.1 cm floor area, 20.2 cm high; MED Associates Inc.). Each chamber contained a food magazine port located 2.5 cm above the floor in the center of the intelligence panel, a red house light located on the wall opposite the food magazine port (on throughout training sessions), and a retractable lever (Med Associates) located 2.5 cm to the left or right of the food receptacle and 6 cm above the floor. This

retractable lever was illuminated when extended with a white LED light placed inside the lever house. For a lever press to be recorded, a force of ~15 g was needed. The pellet dispenser (Med Associates) delivered one 45 mg banana-flavored sucrose pellet (Bio-Serv) into the food magazine port at a time. A head entry was recorded each time a rat broke the infrared photobeam located inside the food magazine port. Each conditioning chamber was placed in a sound-reducing enclosure and a ventilating fan generated background noise. Data collection was controlled by Med-PC IV Behavioral Control Software Suite.

PCA testing and primary measures

One day prior to PCA testing, rats were placed into the test chamber with the red house light illuminated and the lever retracted. 25 food pellets were delivered in accordance with a variable schedule (VI-30; 0-60 s) to foster the reliable retrieval of pellets from the food magazine port. During PCA testing, trials began by illuminating and extending the lever (CS) for 8 s. Immediately following the retraction of the lever, a single banana pellet was delivered into the magazine port (US) and a variable intertrial interval (ITI; 90±60 s) started immediately. An individual PCA training session consisted of 25 trials and lasted 35-40 min. The following 5 measures were extracted from individual trials and collapsed over individual PCA sessions: (1) number of lever deflections (contacts); (2) latency to first lever deflection; (3) number of head entries into the food magazine port (referred to as food cup entries) during the presentation of the CS; (4) latency to the first magazine port entry after CS presentation; (5) number of magazine port entries during the ITI. Data from animals included in the final analysis reflected sessions during which rats consumed all food pellets.

Phenotype classification

The propensity for rats to approach the lever CS versus the food magazine port during the CS period was expressed by a PCA index score. Briefly, the PCA index score consisted of averaging three measures of the conditioned approach on the 4th and 5th day of testing (e.g., Robinson and Flagel, 2009 [\[4\]](#); Meyer et al., 2012 [\[5\]](#); Yager et al., 2015 [\[6\]](#); Pitchers et al., 2017b [\[7\]](#); Pitchers et al., 2017a [\[8\]](#)): (1) the probability of contacting either the lever CS or food magazine port during the CS period ($P(\text{lever}) - P(\text{food port})$); (2) the response bias for contacting the lever CS or the food magazine port during the CS period: $(\# \text{ lever CS contacts} - \# \text{ food magazine port contacts}) / (\# \text{ lever CS contacts} + \# \text{ food magazine port contacts})$; and (3) the latency to contact the lever CS or the food magazine port during the CS period: $(\text{food magazine port contact latency} - \text{lever CS contact latency}) / 8$. Averaging these three measures yields PCA index scores ranging from -1.0 to +1.0, where +1.0 indicates an animal made a sign-tracking CR on every trial and -1.0 a goal-tracking CR on every trial. Further, rats with an averaged PCA index score ranging from -1.0 to -0.5 were classified as GTs, and rats with a PCA index score between +0.5 and +1.0 as STs.

Cue-triggered turning task (CTTT)

Apparatus

The 32.51 cm wide by 153.67 cm long treadmill was constructed by modifying a 2200 Series Flat Belt End Drive Dorner conveyor (Dorner, WI). The conveyor belt surface material was polypropylene, friction-resistant, and easily cleaned with ethanol or soap. The conveyor was paired with a Dorner Variable Speed Controller (Dorner, WI), with speeds ranging from 0 cm/s to 32 cm/s or 19.2 m/min. The Variable Speed Controller included a reversing controller, allowing the belt to rotate clockwise, “forward,” or counterclockwise, “reverse.” An in-house-made Faraday cage was placed on top of this conveyor, 145.10 cm long, 32.51 cm wide, and 38.1 cm tall. The wooden frame of the Faraday cage was enclosed by a woven copper mesh and grounded to block out static electric fields. Plexiglass inserts were placed inside the Faraday cage to prevent rodents from chewing on the mesh. Copper reward ports were installed on either end of the treadmill to

allow delivery of 45 mg banana pellets. Experimenters raised two wooden panels on the cage to place rodents on the treadmill. The 28V DC, 100 mA stimulus light was approximately 2.54 cm in diameter with a flat lens and mounted on both sides lengthwise on the Faraday cage (MedAssociates, Inc., St. Albans, VT). The auditory cue was generated by a Mallory-SonAlert audible device. This 28V DC, 18 mA device was mounted on the center of the Faraday cage and was approximately 4.3 cm long by 4.3 cm wide by 3.6 cm tall. This device emitted a continuous tone at 68 dBA sound pressure level, which is within the acceptable range for chronic presentations, to neither elicit a fearful response nor induce hearing loss (Turner et al., 2005 [↗](#); Castelhamo-Carlos and Baumanns, 2009 [↗](#)). These devices were controlled by a MedAssociate interface and a relay module to run our custom program. Performance sessions were videotaped with four web cameras (Logitech C920x HD Pro Webcam, Full HD 1080p/30fps) and relayed to an Intel Xeon workstation (Dell, Round Rock, TX) via USB cords and processed with OBS Studio (Open Broadcaster Software, free and open-source software).

Training regimen

Rats were trained to walk on a treadmill until the onset of one of two cues (tone or light, presented for 2 s), indicating that following a 5-s treadmill stop, 1 s after cue onset, the treadmill restarted in the reverse or same direction (Avila et al., 2020 [↗](#)). Rats were trained using either the tone as the turn cue and the light as the stop cue, or *vice versa*. Rats learned to respond to turn cues by turning around, and to the stop cue by stopping and not turning. Cued turns and cued stops were rewarded by manually delivering a banana-flavored sucrose pellet (45 mg; Bio-Serv) into the reward port following the treadmill stop.

Rats first underwent a treadmill acclimation regimen using a procedure adapted from Arnold and Salvatore (2014) [↗](#). First, rats were handled for 20 min for two days. On the second day, rats were placed in the Faraday cage for 15 min with the belt paused. All sessions were conducted under dim red lighting to minimize stressors and increase the discriminability of the light cue. For a week, rats were placed into the Faraday cage for 3 min with the belt paused and then trained to walk on the treadmill at speeds up to 9.6 cm/s or approximately 6 m/min. The experimenter manually set the initial rate and direction of the treadmill, with the initial direction counterbalanced within and across subjects. During this training, rats were reoriented by gentle prodding if they began to walk in the opposite direction of the belt. After each session, rats were immediately returned to their home cage and housing room. Between sessions and subjects, the Plexiglass walls of the Faraday cage and the treadmill belt were wiped down with 70% ethanol and soap.

The initial treadmill speed was set at 3.2 cm/s for the first two days and gradually increased by 1.6 cm/s every 2 min. Each session lasted 20 min, so the final walking speed reached 8 cm/s. On the third and fourth days, the initial rate was set at 6.2 cm/s, increasing by 1.6 cm/s every 5 min until the maximum speed of 9.6 cm/s was reached. On the fifth and sixth day, rats walked at 8.0 cm/s for 10 min and 9.6 cm/s for the last 10 min. Finally, on the seventh day of this training regimen, rats walked at 9.6 cm/s for 20 min. In the next training phase, rats were placed on the treadmill and presented with turn and stop cues. The modality of the turn and stop cues was counterbalanced across animals.

Contrary to the acclimation training phase, during which experimenters manually controlled the treadmill, this phase was controlled entirely by custom scripts using Med-PC software and interface (MedAssociates). Rats walked at a speed of 9.6 cm/s and cues were presented for 2 s. Maximally two successive presentations of a cue of the same modality were allowed. The intertrial interval (ITI) was 60±30 s. The total number of trials per daily session was 18, with a maximum duration of a test session of 20 min. Rats underwent cue training for approximately two to three weeks. Then, upon reaching the performance criterion, defined as 70% correct responses to either cue for two consecutive days, rats were tested and videotaped for four additional days. Data from

these four sessions were used to determine potential phenotype and sex-based differences in baseline performance. Thereafter, rats underwent surgery for chronic microelectrode array implantation.

CTTT performance measures

Performance measures were cued turns, missed turns, cued stops, and false turns, and extracted from offline scoring of session videos. For a successful turn to be scored, the animal must have initiated a turn, defined by a rotation of the longitudinal orientation of the body of at least 90°, before the treadmill restart, that is, within 8 s of the cue onset. A stop was defined as a cessation of forward movement. For stops occurring before the treadmill stopped, rats would typically stop while positioned at the backend of the treadmill so that following a stop, the treadmill would transport them to the front of the treadmill (within the remaining 1-3 s until the treadmill stopped), without the rat contacting the front end of the test chamber. Following the treadmill stop and during the 5-s pause, a false turn was scored if the animal (falsely) turned. The time of initiation and completion of cued turns, relative to the onset of the cue, were extracted from session videos. For the analysis of baseline CTTT performance, and because rats generated a variable number of cued turns during the 4 sessions used for this analysis, individual turn onset and completion times were averaged across the test sessions. Additional performance measures were extracted from training and testing sessions to explore relationships between cued turning and stopping performance, phenotype and, subsequently, glutamatergic transients (detailed in Results).

Amperometric recordings of analysis of glutamate currents

Electrode preparation and calibration

Extracellular glutamate concentrations were measured on electrode surfaces featuring immobilized glutamate oxidase (GO) and by performing fixed-potential amperometry to determine hydrogen peroxide concentrations resulting from GO-catalyzed oxidation of glutamate. Based on calibration curves generated *in vitro*, the resulting currents were expressed as micromolar glutamate concentrations at the recording site (Burmeister and Gerhardt, 2001 [DOI](#); Rutherford et al., 2007 [DOI](#); Hascup et al., 2008 [DOI](#); Parikh et al., 2008 [DOI](#); Parikh et al., 2010a [DOI](#); Parikh et al., 2014 [DOI](#); Clay and Monbouquette, 2018 [DOI](#); Bermingham et al., 2022 [DOI](#)). The configuration consisted of a ceramic backbone probe (Quanteon LLC, Nicholasville, KY, USA) housing four recording sites, each 15×333 μm, crafted from platinum-iridium (Pt/Ir). These recording sites were grouped into two pairs, separated by 30 μm between each member of a pair and a 100 μm vertical spacing between the pairs, so that the two pairs of electrodes were linearly arranged along the shank of the probe. Microelectrode arrays were modified for *in vivo* recordings in behaving rats by soldering four enamel-coated magnet wires (30 ga) to the terminals on the electrode panel and the other end to gold-pin connectors. Reference electrodes were constructed by soldering Ag/AgCl reference electrodes prepared from 0.008" silver wire (A-M Systems, Carlsberg, WA) to gold-pin connectors. The pins were then inserted into a 9-pin ABS plug (GS09PLG-220, Grinder Scientific) and adhered to the microelectrode with epoxy. Custom 9-pin ABS plugs were also printed with a STUDIO G2 3D printer (BigRep, Berlin, Germany). The top pair of electrodes, termed active sites, were coated with recombinant L-GO (US Biological Life Sciences) solution (1 U in 1 μL DiH₂O), cross-linked with a BSA and glutaraldehyde mixture, by manually applying microdroplets using a 1-μL Hamilton syringe. The lower pair of electrodes, termed sentinels, were coated with only the BSA and glutaraldehyde mixture, to record currents unrelated to glutamate concentrations. After a minimum 24-hr incubation period to ensure optimal adherence of the enzyme layer, an exclusion layer composed of *meta*-(1,3)-phenylenediamine (*mPD*) (Mitchell, 2004 [DOI](#)) was electroplated onto each recording site's surface by 5-min application of 0.85 V versus an Ag/AgCl reference electrode (Bioanalytical Systems). This exclusion layer prevents sensing electroactive interferents such as ascorbic acid (AA) and catecholamines (e.g., Mitchell, 2004 [DOI](#)). After another 24-hr incubation

period for the electrode sites to dry, recording sites were calibrated to determine the sensitivity for glutamate, selectivity for glutamate versus interferents, stability, and limit of detection of glutamate. Calibrations were conducted using a FAST-16 electrochemical system (Quanteon). A constant voltage of 0.7 V was applied versus an Ag/AgCl reference electrode, and the system was placed in a heated 40 mL bath of 0.05 M PBS. Following a 30-minute baseline recording, aliquots of stock solutions of ascorbic acid (AA; 20 mM), glutamate (20 mM), and dopamine (DA; 2 mM) were added to the calibration beaker such that the final concentrations were 250 μ M AA, 20, 40, and 60 μ M glutamate, and 2 μ M dopamine (see also Wassum et al., 2012 [↗](#); Malvaez et al., 2015 [↗](#)). Changes in amperometric current at individual electrode sites were measured after each solution to calculate the slope (sensitivity), the limit of detection, selectivity for AA and dopamine, and linearity (R^2). Glutamate sensors were required, at a minimum, to have a sensitivity of >5 pA/ μ M glutamate, a limit of detection <1.0 μ M glutamate, glutamate:AA selectivity ratio $>50:1$, minimal changes in current across all channels following DA addition (<3 pA), and a linear response to increasing glutamate concentrations (20–80 μ M glutamate) of $R > 0.95$. The characteristics of the electrodes used in the present experiments exceeded these minimum requirements (Table 1 [↗](#)).

Chronic implantation of microelectrode arrays (MEAs)

Upon having reached stable, criterion-level performance CTTT ($\geq 70\%$ cued turns and stops for two consecutive days), microelectrode arrays were chronically implanted into the DMS. Rats were anesthetized using isoflurane gas (5% induction and 1–3% maintenance) and mounted on a stereotaxic frame on a heating pad to maintain a 37°C body temperature. Ophthalmic ointment lubricated eyes. A craniotomy and durotomy using a bent 27-gauge needle for dura removal were performed above the right DMS (AP: +0.50 mm; ML: –2.20 mm from bregma). Three stainless steel screws were threaded into the cranium. The MEA was then lowered 4.5 mm dorsoventrally from the dura into the striatum. At the same time, an Ag/AgCl reference electrode was implanted at a remote site in the contralateral hemisphere. The microelectrode assembly was anchored with methyl methacrylate dental cement, and exposed regions of the skull were filled with a translucent, medium-viscosity silicone adhesive to minimize leakage of the dental cement onto the brain. Animals rested over a 48-hour recovery period before moving to the next phase of the experiment. Amperometric recordings were collected by connecting the head stages to a FAST-16 potentiostat/data electrochemical system (Quanteon) via a shielded cable and low-impedance commutator. The hydrogen peroxide by-product of GO catalyzation was electrochemically oxidized by applying 0.7 V versus the Ag/AgCl reference electrode and digitized at a sampling rate of 5 Hz. Behavioral sessions began following a 50-minute baseline recording period. Consistent with prior evidence showing that implanted MEAs maintain stable sensitivity and reliability for at least seven days after implantation, electrochemical recordings were completed within seven days of MEA implantation (Rutherford et al., 2007 [↗](#)).

Amperometry data processing and analysis of glutamate peaks

Baseline correction and normalization. Electrochemical recording data were processed using a custom MATLAB (MathWorks) script. The background current recorded on sentinel sites was subtracted from the current recorded on the active sites, normalized by the response to dopamine (Burmeister and Gerhardt, 2001 [↗](#)), and then converted to glutamate concentration based on calibration curves. Normalization of net currents by dopamine currents corrected for variations in the efficacy of the mPD barrier (Burmeister and Gerhardt, 2001 [↗](#)). Such normalization was only computed for recordings with electrodes which responded to dopamine (3 out of a total of 15 electrodes; see Table 1 [↗](#)). Trial-associated glutamate concentrations were determined for three periods:

	Sensitivity (Slope) to Glutamate (M±SEM)	Selectivity for Glutamate over Ascorbic Acid (AA; M±SEM)	Limit of Detection of Glutamate M±SEM)	Linearity of Response (R ² , M±SEM)	Dopamine-Evoked Current (individual electrode values ¹)
Minimum Criteria	>5 pA/μM	>50:1 glutamate:AA	<1.0 μM	R ² >0.95	n/a
Measured (N=15 electrodes; 8 in GTs, 7 in STs)	12.0±0.2	113.1±23.5	0.3±0	0.9±0	0.26, 0.47 and 0.53 pA ¹

¹ Recordings of glutamate reflecting currents *in vivo* using these three electrodes (2 in GTs and 1 in a ST) were normalized (divided) by dopamine-evoked currents.

Table 1.

Electrode *in vitro* calibration characteristics

1. Baseline period: The concentration of glutamate currents recorded over 2.5 s prior to cue onset served as a baseline for determining the number and levels of task event-locked peaks. Therefore, glutamate concentrations shown in graphs depict sentinel-corrected, normalized (where applicable) and baseline-corrected levels. Across trials and entire sessions, only minimal drift of baseline glutamate levels was observed; however, baseline-based corrections further reduced the impact of potential minute-based variations in glutamate concentrations.
2. Cue presentation period: Glutamate concentrations recorded throughout the 2-s presentation of cues were used to determine cue-locked peak characteristics.
3. Reward delivery period: Currents recorded during a 2-s period following reward delivery were used to determine reward delivery-locked peaks.

The analysis of event-locked glutamate concentrations focused on determining maximum peak concentrations, the number of glutamate peaks, and the timing of the first peak relative to an event. Prior studies on the effects of depolarization of synaptic terminals, blocking such depolarization, or of pharmacological manipulations of the excitability of terminals, confirmed that peak concentrations of extracellular glutamate indicate the degree and extent of terminal depolarization (Hascup et al., 2008 [↗](#); Parikh et al., 2008 [↗](#); Parikh et al., 2010b [↗](#); Mattinson et al., 2011 [↗](#); Quintero et al., 2011 [↗](#); Parikh et al., 2014 [↗](#)).

Peak identification criteria. The PeakDet function in MATLAB (MathWorks) was utilized to identify peaks, requiring setting threshold and minimum differences.

1. Threshold for peak identification: To be identified as a peak, glutamate concentrations needed to be 3 standard deviations (SD) above the average baseline current (recorded over 2.5 s before cue presentation).
2. Minimum differences: For a value to be identified as a peak, the preceding and subsequent values needed to be at least 1 SD (derived from baseline data) below the peak value.
3. Adjacent points: If multiple adjacent points were above 3 SDs of the baseline but within 1 SD from each other, the highest of these points was counted as a peak. In the absence of this additional criterion for determining peaks, none of such multiple, adjacent peaks would be counted, as none would have been bordered on both sides by points 1 SD below peak value (second criterion). Furthermore, applying this third criterion to such situations limited an escalation of peak counts during relatively sustained elevations of currents. Only 41 out of 752 traces (5.5%) required the application of this third criterion.

These criteria for identifying peaks were adopted from peak analyses employed in neurophysiological studies (e.g., Ghosh et al., 2022 [↗](#)) and ensured a robust separation of peaks from potentially noisy recordings of relatively small currents and the false identification of peaks during fluctuations of relatively higher currents.

The maximum peak concentration (μM), time to peak (s), and number of peaks were extracted from each trace used for the final analyses. Peak amplitude was defined as the highest glutamate concentration reached within 2 s from the onset of a cue or from reward delivery. The time to peak was defined as the time from cue onset or reward delivery to the first peak. During missed turns, peaks were not identified in a small fraction of trials (<2% in both phenotypes), yielding missed data for this measure. The number of peaks were counted across a 2-s period from cue onset or reward delivery.

Trace inclusion and exclusion criteria. For each response category (cued turns, misses, cued stops, false turns), glutamate traces were included into the final analyses if; a) if the electrode met the *in vitro* calibration criteria, b) traces were devoid of major electrostatic interferences (such as resulting from the headstage contacting the reward port); c) task compliance was apparent

(animals walking continuously counter the direction of the treadmill and promptly reversing direction upon cue or treadmill onset); d) the accuracy of the electrode placement was confirmed, determined following the completion of experiments. The data from one rat that developed seizures after the 2nd day of recordings were completely excluded from the final analyses. As a result, 1-27 (range) traces per rat and response category were included in the final analyses.

Verification of MEA placements in the DMS

MEAs were chronically implanted into the DMS to record extracellular glutamate levels while performing the CTTT. Following the completion of experiments, administration of a lethal dose of sodium pentobarbital (270 mg/kg, i.p.) was followed by transcardial perfusion of saline, followed by 4% paraformaldehyde in 0.15 M sodium-phosphate solution, pH 7.4. Extracted brains were postfixed in 4% paraformaldehyde for 24 hrs, then submerged in 30% sucrose solution until they sank. Using a freezing microtome (CM 2000R; Leica), 35- μ m thick brain slices were sectioned and stored in cryoprotectant until further histologic processing. Sections were mounted and processed with a Cresyl Violet Nissl stain to verify placements. A Leica DM400B digital microscope was used to photomicrograph the sections at 1.25X and 5X magnification at three A-P levels (0.2 mm, 0.5 mm, and 1.0 mm).

Intracranial infusions of a Cre-dependent DREADD and a retrograde Cre-vector

We utilized a pathway-specific dual-vector chemogenetic strategy (e.g., Sherafat et al., 2020 [DOI](#)) to selectively inhibit the activity of fronto-cortical projections to the DMS. Intracranial infusions of a Cre-dependent Designer Receptor Activated Only by Designer Drug (DREADD) and a retrogradely transported Cre-expressing plasmid were carried out following PCA screening and prior to CTTT acquisition training. In a subset of rats, a second surgery, carried out following the acquisition of the CTTT, was conducted to implant MEAs into the DMS. The viral vectors containing the Cre-dependent plasmid pAAV-hSyn-DIO-hM4D(Gi)-mCherry (AddGene #44362-AAV8; titer of 2.1×10^{13} GC/mL), Cre-dependent control plasmid pAAV-hSyn-DIO-mCherry (AddGene #50459-AAV8; titer of 2.2×10^{13} GC/mL), or the retrogradely transported Cre-expressing plasmid pENN-rAAV-hSyn-HI-eGFP-Cre-WPRE-SV40 #105540-AAVrg; titer of 1.9×10^{13} GC/mL) were infused into the lower layers of the prelimbic cortex (for the anatomical organization of fronto-striatal projections see, e.g., Maily et al., 2013 [DOI](#)) and the DMS, respectively.

Craniotomies and durotomies were carefully performed above four sites of the DMS using a bent 27-gauge needle for dura removal to minimize the impact on the subsequent implantation of recording electrodes and electrochemical recordings. One μ L of pENN-rAAV-hSyn-HI-eGFP-Cre-WPRE-SV40 vector was infused (bolus) into the DMS at two sites per hemisphere (AP: +0.2/1.2; ML: $\pm 2.5/2.2$; DV: -4.5 mm from dura) to retrogradely transfect afferent projections. In addition, one μ L of AAV-hSyn-DIO-hM4D(Gi)-mCherry or AAV-hSyn-DIO-mCherry (control vector) was infused (bolus) into the prelimbic cortex (AP: +3.2; ML: ± 0.7 ; DV: -3.5 mm from dura) to allow for the selective inhibition of cortico-striatal projections. The injector was left in place for 8 min to minimize diffusion into the injector tract. Rodents recovered for one week before treadmill and CTTT training.

Clozapine N-oxide (CNO)

CNO was obtained from Tocris Bioscience (Bristol, United Kingdom) and dissolved 10 mg/ml in 6% DMSO in 0.9% NaCl solution. CNO (5.0 mg/ml/kg; i.p.) or vehicle was administered i.p. 50 min before the onset of CTTT testing. In rats also equipped for electrochemical recording of dorsomedial glutamate levels, CNO was given just prior to the onset of recording the pre-task baseline (above). Rats were given vehicle and CNO on alternate days. Regarding potential off-target effects of clozapine (Gomez et al., 2017 [DOI](#)), we and others have previously consistently failed to detect effects of this dose of CNO in rats expressing the control vector, including in rats

performing the CTTT, complex movement control tasks, or an operant sustained attention task (Jendryka et al., 2019 [↗](#); Avila et al., 2020 [↗](#); Kucinski et al., 2022 [↗](#)). Given effective dose ranges of clozapine in rodents performing complex behavioral tasks (Martinez and Sarter, 2008 [↗](#)), and given the proposed conversion rate of CNO to clozapine, an approximately 50-100-fold higher dose of clozapine would be required to produce significant effects (see also Mahler and Aston-Jones, 2018 [↗](#); Lawson et al., 2023 [↗](#)).

Visualization and quantification of GFP/mCherry-expressing neurons

We amplified the mCherry fluorescent reporter signal of the inhibitory hM4Di DREADD vector to enhance the evaluation of the transfection efficacy and distribution of neurons co-expressing mCherry and eGFP. The eGFP fluorescent label did not necessitate signal enhancement. Sections underwent six washes for 5 min each in 0.1 M PBS, pH 7.3, and then were immersed in 0.1% Triton X-100 diluted in PBS for 15 min. After three 5-min PBS rinses, sections incubated for 60 min at room temperature (RT) in the blocking solution, 1% normal donkey serum (NDS) and 1% Triton X-100 made in PBS. Sections were incubated overnight in the primary antibody (rabbit anti-mCherry, ab167453, Abcam; 1:500; diluted in blocking solution to prevent non-specific binding). The next day, and following three 5-min PBS rinses, sections were incubated for 90 min at RT in the secondary antibody (donkey anti-rabbit conjugated to Alexa 594, PIA32754, Invitrogen, 1:500). Following three 5 min-rinses with PBS and sections were mounted, air dried, and cover-slipped with Vectashield Antifade Mounting Medium (H-1000; Vector Laboratories).

A Zeiss LM 700 confocal microscope, equipped for sequential multi-track acquisition with 488- and 561nm excitation lines, along with specific filter sets for Alexa 488 (Zeiss filter set 38 HE) and Alexa 594 (Zeiss filter set 54 HE), was employed for visualizing and capturing images of fluorescent neurons. Images were taken at magnifications of 10x to confirm placements of infusions in the DMS, 20x and 40x at five anterior-posterior levels (frontal cortex: 3.0 mm and 3.24 mm, striatum: 1.2 mm, 0.5 mm, and 0.2 mm) for verification and documentation of single and double-labeled cells in cortex and striatum.

We employed a modification of a prior, semi-quantitative estimation of transfection efficacy and space (Avila et al., 2020 [↗](#)). Briefly, GFP labeling that was centered in the striatal target region of prelimbic projections (Mailly et al., 2013 [↗](#)) was assigned the highest transfection score (5), whereas labeling that extended into more lateral and ventral regions and only partially covered the target region were assigned lower scores. Likewise, mCherry labeling that exclusively covered the entire prelimbic cortex was assigned the highest score. Partial expression of mCherry in this region, or labeling that extended dorsally or ventrally beyond the prelimbic cortex, were assigned lower transfection efficacy scores. Furthermore, single- and double-labeled neurons were counted in the prelimbic region, and the proportion of double-labeled neurons was computed for each rat. Counting frames were superimposed on each subregion using the ImageJ multipoint tool. Cells were counted on two sections per brain, yielding four counts per brain and subregion. Averaged counts obtained from each brain and subregion were used for further analyses. To produce images of neurons expressing mCherry and GFP, and control for potential “bleed-through” (e.g., North, 2006 [↗](#)), we used the split view feature within Zen Black software (ZEN 2.31 SP1 Black Edition; Zeiss) for examination of individual channels with multi-track images. Tile scanning/stitching techniques were implemented at both 10x (3 × 2 tiles, covering an area of 3455.94 by 2370.64 μm) and 20x magnifications (4 × 3 tiles, covering an area of 1407.88 by 1135.31 μm).

Experimental design and statistical analyses

Behavioral data

Chi-square tests were used to determine the effects of sex and the commercial source of rats (vendor) on the distribution of PCA screening scores. Baseline CTTT performance was analyzed using conventional repeated measures ANOVAs; associated graphs depict individual values, means, and 95% Confidence Intervals (CI). Conventional ANOVAs were computed using SPSS for Windows (version 28.0; SPSS) and GraphPad Prism (version 10.1.2).

Glutamate concentrations and effects of CNO

Extracellular glutamate concentrations were sentinel-corrected, normalized to dopamine (if applicable), and pre-cue baseline corrected (see above for details). From these data, peak concentrations and peak timing information were extracted (see above for the definition of the three measures and of peaks). Given the complexity of the experimental design (variable number of rats per phenotype, turn and stop trials, and of cue-evoked glutamatergic transients), here we first determined potential phenotype-specific differences in glutamate peaks across trial types and response categories, followed, in cases, by *post hoc* analyses of event-related glutamate levels within individual phenotypes. Linear-mixed effects models (LMMs) with restricted maximum likelihood estimation were computed in SPSS for Windows (version 28.0; SPSS). A key reason for using LMMs concerned the definition of sample size (number of animals versus the number of individual traces collected from each animal). Moreover, sample sizes may vary across repeated or otherwise dependent measures (e.g., Schielzeth et al., 2020 [DOI](#); Yu et al., 2022 [DOI](#)). Data from repeated sessions were used to compute repeated fixed effects, and phenotype was used to determine between-subjects fixed effects, with a subject identifier random intercept. Separate LMMs were computed for the analysis of glutamate levels during four response categories (cued turns, missed turns, cued stops, reward delivery); the three measures analyzed per response categories (maximum peak levels, time to first peak, and number of peaks; defined above) were treated as dependent measures. The covariance structures with the lowest Akaike's information criterion were selected for each model (Verbeke and Molenberghs, 2009 [DOI](#)). Main effects of phenotype derived from the analysis of the three measures were followed up using Bonferroni's method for pairwise comparison of the means. When LMMs reported significant interactions, repeated measures ANOVAs were used to compare data from repeated sessions, with Huynh-Feldt-corrected *F* values and corrected degrees of freedom applied in case of violation of the sphericity assumption. Data graphs showing LMM-analyzed glutamate concentrations depict individual values, estimated marginal means (EMMs), and 95% Confidence Intervals; CI).

Contingency table analyses

Glutamate trace characteristics were extracted from 548 recordings of turn cue trials, 364 of which yielded a turn (GTs: 206, STs: 158) and 184 a miss (GTs: 112, STs: 72). Contingent on threshold maximum peak concentrations, and the presence of a single or multiple cue-evoked glutamate peaks, the proportions of turns and misses were compared using contingency table analyses (*P* values were computed using Fisher's exact test; GraphPad Prism). Furthermore, the relative probability for turns in GTs, as well as the reciprocal value, were derived from these analyses (Koopman asymptotic score; Koopman, 1984 [DOI](#); Motulsky, 2018 [DOI](#)).

Effects of CNO on CTTT performance

The effects of CNO or its vehicle on cued turns and stops, in rats expressing the inhibitory DREADD or the empty control vector, were analyzed using repeated measures ANOVA (general linear model) on the effects of phenotype and treatment day. For clarity, effects on the two behavioral measures were analyzed separately, at $\alpha=0.05/2$. Tukey's Honest Significant Difference test was used to compare, *post hoc*, the effects of the 1st and 2nd administration of CNO with the effects of the 1st and 2nd administration of vehicle (GraphPad Prism).

P values and effect sizes

Exact *P* values were reported (Greenwald et al., 1996 [DOI](#); Sarter and Fritschy, 2008 [DOI](#); Michel et al., 2020 [DOI](#)). For key findings, effect sizes were computed using generalized eta squared (η^2) (effect sizes of 0.02 typically are classified as small, 0.13 as medium, and 0.26 as large; e.g., Bakeman, 2005 [DOI](#)), derived from Bonferroni pairwise comparisons of estimated marginal means (Greenwald et al., 1996 [DOI](#)).

Results

Phenotype screening and distribution by sex and vendor

PCA screening generated five behavioral measures indicating the speed and frequency of contacting the lever (Pavlovian CS) versus the speed and frequency of head entries into the food port. These measures were collapsed into the PCA score, indicating the degree of sign-tracking and goal-tracking, respectively, of individual rats over 5 PCA test sessions. Prior analyses have consistently shown that all rats orient toward the CS, that is, all rats learn the predictive significance of the CS, but only some – the STs – approach and contact the CS (**Fig. 1a** [DOI](#), top right photo), while GTs do not contact the CS but approach and enter the food port (Fig., 1a, bottom right photo; e.g., Robinson and Flagel, 2009 [DOI](#); Meyer et al., 2012 [DOI](#); Pitchers et al., 2017b [DOI](#)).

PCA scores from the last two sessions were averaged and used to classify rats as STs and GTs. PCA screening of *N*=378 (215 females) rats yielded 113 GTs (30%), 155 rats with intermediate scores (INs; 41%), and 110 STs (29%). **Figure 1a** [DOI](#) shows the distribution of PCA scores across the five test sessions of the rats used for CTTT training and amperometric recordings (79 rats GTs, 13 females; 22 STs, 13 females; data from rats with intermediate PCA scores are not shown).

As PCA screening was conducted in four separate cohorts of rats, separated by 2–19 months, we first determined that PCA scores did not differ across these cohorts (no main effects of cohort on response bias (respbias), probability difference (probdiff), latency, or PCA index; all *F*<1.74, all *P*>0.16). Chi-square tests were used to determine if sex or the commercial source of the rats (vendor) influenced the distribution of phenotypes. Consistent with prior reports (Pitchers et al., 2015 [DOI](#)), the distribution of PCA scores did not differ by sex ($X^2(2, N=378)=0.35, P=0.56$; **Fig. 1b** [DOI](#)). However, the distribution of PCA scores from rats obtained from the two vendors differed significantly ($X^2(2, N=378)=36.62, P<0.001$; **Fig. 1c** [DOI](#); note that the percentages of phenotypes shown in **Fig. 1c** [DOI](#) were calculated individually for each vendor). Relatively more rats obtained from Inotiv displayed lever-directed behaviors, while rats supplied by Taconic relatively more frequently exhibited goal cup-directed behaviors. A follow-up analysis rejected the possibility that vendor-specific PCA score distribution differed by sex (Inotiv: $X^2(2, N=133)=1.30, P=0.52$; Taconic: $X^2(2, N=245)=1.41, P=0.50$). GTs and STs of both sexes and from either vendor were combined for use in subsequent experiments.

CTTT acquisition and criterion performance

Upon reaching criterion performance in the CTTT (see the apparatus, illustration of task rules and of the events and timeline constituting a trial in **Fig. 2** [DOI](#)), defined as >70% cued turns and cued stops for two consecutive days/sessions, and prior to the intracranial implantation of microelectrode arrays, the relative number of cued turns and stops were recorded from an additional 4 test sessions to determine baseline CTTT performance of GTs and STs.

The number of training sessions required by GTs (*n*=29, 13 females) and STs (*n*=22, 12 females) to reach CTTT criterion performance, defined as 70% correct responses to either cue for two consecutive sessions, did not differ significantly (*F*(1,47)=0.01, *P*=0.94; **Fig. 3a** [DOI](#)). Likewise, male

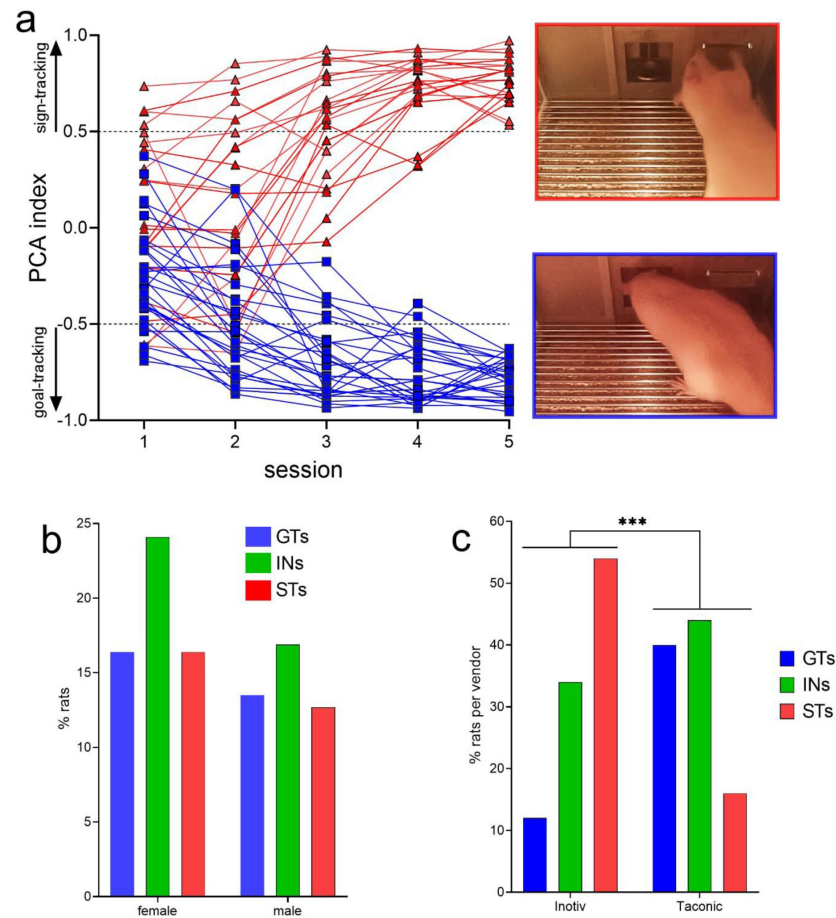


Figure 1

a Distribution of PCA scores of rats eventually categorized as GTs or STs. A total of N=378 rats (215 females) underwent PCA screening. Final phenotypic classification was based on PCA score averages from the 4th and 5th test session (rats with intermediate PCA scores, INs, are not shown). The photos on the right depict lever (CS) contact, indicative of sign-tracking behavior (top) versus CS-triggered food port entry, signifying goal-tracking behavior (bottom). The distribution of PCA scores was unaffected by sex (**b**) but differed among rats obtained from the two vendors (**c**). Relatively more rats from Inotiv were classified as STs while rats from Taconic tended to emerge relatively more often as GTs (note that in **c** phenotype percentages are expressed based on the total number of rats per vendor). The distribution of vendor-specific PCA scores was not affected by sex (see Results; not shown).

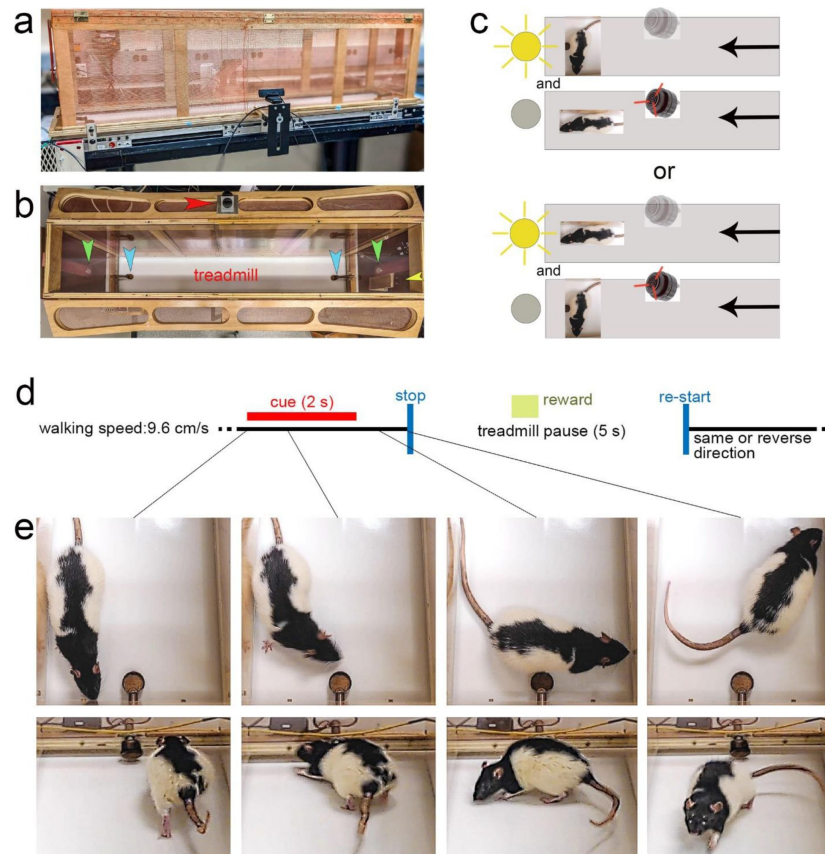


Figure 2

Illustration of the CTTT, task rules, trial timeline, and of a cued turn. **a** shows a lateral view of 1.5-m long CTTT, with Faraday shielding comprised of copper wire mesh surrounding the motor and wooden frame of the Plexiglas-lined enclosure (both grounded), 2 mounted cue lights on both ends, 1 mounted audible device centered (not visible here), and a copper reward port installed on either end of the treadmill. Web cameras were placed on each side of the treadmill for off-line analysis of the rat's performance. **b**: Top-down view of the treadmill apparatus showing the position of the stimulus lights (green arrows), the SonAlert device to generate the tone (red arrow), the two copper reward ports (blue arrows), and a camera situated outside of the enclosure (yellow arrow). As illustrated in **c**, half of the rats were trained to turn in response to the light stimulus and to stop in response to the tone (upper two panels), and the other half acquired the reversed stimulus-response rules (lower 2 panels; note that in this example the treadmill moves from right to left – see arrows on the right – and, in case of turn cue would restart after the pause – see **d** – moving from left to right). **d** Trial events and timeline. The CTTT required rats to walk on a treadmill (walking speed: 9.6 cm/s). Presentation of a turn or stop cue for 2 s indicated that the treadmill stopped 1 s later, and restarted after a 5-s pause, in the reverse or same direction, respectively. Rats were trained using either the tone as the turn cue and the light as the stop cue, or vice versa. Successful turns and stops without turns, respectively, were rewarded by delivering a 45 mg banana pellet, on average 3.6 s following cue offset (see green square in **d**). **e** shows photographs of a cued turn, initiated within the first second of cue onset (see lines connecting the individual photos to the task timeline in **d**; the upper four photographs were extracted from a top-down video and the lower four from a video taken using a camera situated at the backside of the apparatus (yellow arrow in **b**), facing the rat following completion of the turn). Successive presentation of cues was separated by a variable intertrial interval (ITI) of 60 ± 45 s during which the rat continued walking the treadmill.

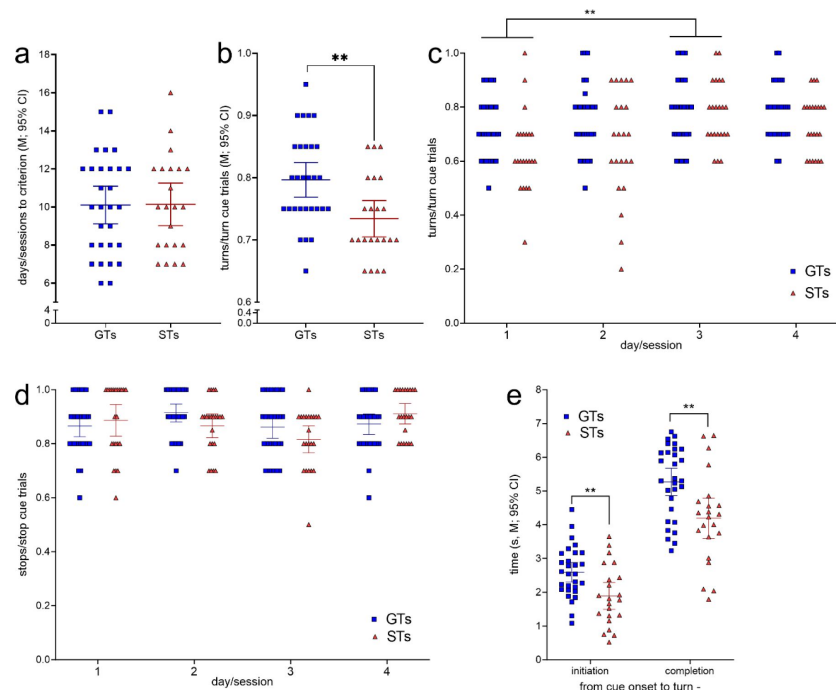


Figure 3

CTTT acquisition and asymptotic performance by GTs (n=29, 13 females) and STs (n=22, 12 females). **a** The number of training sessions to reach CTTT criterion performance, defined as 70% correct responses to either cue for two consecutive days, neither differed between the phenotypes nor the sexes. However, baseline performance, based on data from four test sessions conducted after rats had reached criterion performance, indicated that GTs scored more cued turns than STs (main effect of phenotype; **b**). Moreover, a main effect of day reflected that rats performed more cued turns on day 3 than on day 1 (*post hoc* Bonferroni test; **c**; note that **c** and **d** show response ratios but ANOVAs were conducted using arcsine transformed data because ratio data violated homoscedasticity). **d** The relative number of cued stops did not differ between the phenotypes. There were no main effects of sex and no significant interactions between the effects of phenotype, day and sex on either measure. **e** Relative to the time of cue onset, GTs initiated and completed cued turns later than STs. Reflecting the parallel effects of phenotype on initiation and completion times, the actual time needed to complete turns did not differ between the phenotypes (not shown; *, **, ***: $P < 0.05$, 0.01, 0.001).

and female rats acquired this task at comparable rates (main effect of sex: $F(1,47)=0.81$, $P=0.37$; phenotype x sex: $F(1,47)=0.27$, $P=0.61$).

After having reached performance criterion, GTs scored significantly more cued turns than STs across four subsequent test sessions (main effect of phenotype: $F(1,47)=5.03$, $P=0.003$, $\eta_p^2=0.097$; **Fig. 3b**). A main effect of day (or session; $F(3,141)=4.96$, $P=0.003$, $\eta_p^2=0.096$) reflected that rats generated more cued turns during the third when compared with the first of the four sessions used for the determination of CTTT baseline performance (**Fig. 3c**; day 1: 0.7 (mean) cued turns/trials; day 3: 0.8). Sex did not affect turning rates ($F(1,47)=0.01$, $P=0.95$) and interactions between phenotype, sex and day remained insignificant (all $F<2.10$, all $P>0.15$). The proportion of cued stops following stop cue presentation did not differ by phenotype, day or sex (all main effects and interactions: all $F<3.06$; all $P>0.052$; **Fig. 3d**).

Relative to the time of cue onset, GTs initiated and completed cued turns significantly later than STs (main effects of phenotype; initiation: $F(1,51)=8.84$, $P=0.005$, $\eta^2=0.16$; completion: $F(1,51)=9.40$, $p=0.004$, $\eta^2=0.17$; **Fig. 3e**; effects of sex and interactions between the effects of sex and phenotype: all $F<1.03$, all $P>0.32$). Reflecting the parallel effects of phenotype on initiation and completion time relative to cue onset, the time needed to complete turns, that is, the time from turn initiation to completion, did not differ between the phenotypes (main effects of phenotype, sex, and interactions: all $F<3.51$, all $P>0.07$); GTs: 2.7 ± 0.2 s (M, SEM); STs: 2.3 ± 0.2 s; not shown).

Taken together, GTs and STs acquired the CTTT within a similar number of training sessions, GTs scored more cued turns than STs, and GTs initiated and completed cued turns, relative to cue onset, at significantly later times relative to cue onset. Subsequent experiments recorded DMS glutamatergic transients during cued turns, cued stops, missed turns, false turns, locked to reward delivery, and tested the role of fronto-striatal glutamatergic projections in generating performance-associated glutamatergic transients. Because of the absence of significant effects of sex, and of interactions involving sex as a factor, in the analysis of baseline CTTT performance, subsequent statistical analyses, using linear mixed effect models to account for effects of animals and variable numbers of transients recorded from each animal, did not further involve sex as a statistical variable. Furthermore, cue modality did not impact turn or stop rates and thus, likewise, was not included as a factor in the final analyses.

Turn and stop cue-locked, phenotype-specific glutamate dynamics

Rats were screened for the classification of the phenotypes, underwent CTTT training until they performed at criterion level, followed by implantation of an MEA into the DMS and subsequent recordings of glutamate currents during CTTT performance. **Figure 4** illustrates the four recording sites fabricated onto a ceramic backbone (**Fig. 4a**), the preparation of pairs of recording sites for the measurement of glutamate concentrations and potential electroactive interferences, and the measurement scheme (**Figs 4b,c**). Following calibration of GO-coated and sentinel electrodes *in vitro* (**Table 1**; a representative example is shown in **Fig. 4d**), MEAs were permanently implanted into the DMS (**Fig. 4e**).

Figures 5a and b show representative glutamate currents recorded during cued turns in GTs and STs. These traces illustrate the predominance of single, turn cue-locked peaks in GTs, contrasting with the more frequent presence of multiple (2 or 3) peaks with relatively smaller maximum amplitudes in STs. Furthermore, reward delivery-locked peaks occurred more reliably, and with relatively higher maximum amplitudes, in STs than in GTs. The focus on glutamate peak characteristics was guided by the view that such peaks reflect the orchestrated yet asynchronous depolarization of glutamatergic terminals, over several hundreds of milliseconds and within fractions of micrometers from the electrode surface, and likely sufficient to stimulate synaptic and extra-synaptic, relatively low-affinity glutamate receptors (Hascup et al., 2008; Parikh et al.,

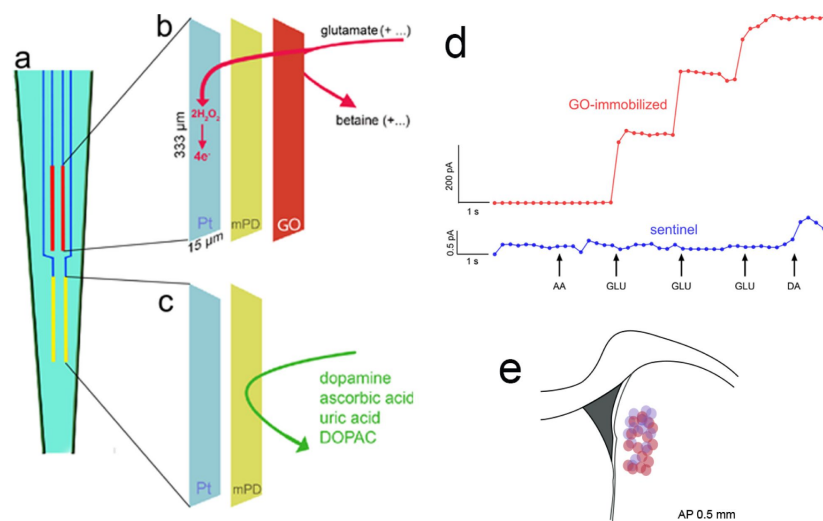


Figure 4

Electrochemical measurement scheme, electrode calibration and placement. **a** Schematic illustration of a ceramic backbone with four platinum-iridium (Pt/Ir) recording sites, organized in two pairs, with each recording site measuring 333 μm long and 15 μm wide. The upper pair was fabricated to measure currents reflecting glutamate concentration (red, also **b**), while the lower pair served as sentinels for background current recording and subtraction (yellow, also **c**). Glutamate oxidase (GO) was immobilized onto the upper (**b**), but not lower (**c**), pair of recording sites. After GO coating, a non-conducting polymer, *m*-(1,3)-phenylenediamine (*m*PD), was electroplated onto both sites to prevent the transfer of small electroactive organic molecules to the Pt sites. Application of a 0.7 V constant potential versus an Ag/AgCl reference electrode (not shown) served to oxidize hydrogen peroxide at individual recording sites (e.g., Burmeister et al., 2000; Parikh et al., 2010b), yielding current proportional to the concentration of glutamate at the recording site. **d** *In vitro* calibration (representative example) indicated the sensitivity, linearity of the response to increasing concentrations of glutamate, and selectivity of GO-coated recording sites (red; see Table 1 for calibration data). The current recorded via the sentinel site did not respond to glutamate and the *m*PD barrier completely prevented ascorbic acid (AA) from generating current. Addition of dopamine (DA) increased the current by about 0.5 pA (note that the ordinate used to graph the current obtained via the GO-immobilized site (200 pA) obscures the visualization of such a small increase). For the 3/15 electrodes that exhibited a dopamine response (>0.1 pA; see Table 1), net currents were normalized by the dopamine response (Burmeister and Gerhardt, 2001). **e** Electrodes were implanted into the DMS; the circles in **e** indicate the location of microelectrode tips in this area (red circles: placements in rats which were recorded during CTTT performance; purple circles: placements in rats in which fronto-striatal projections were silenced).

2008 [Parikh et al., 2010b](#) [Mattinson et al., 2011](#) [Quintero et al., 2011](#) [Parikh et al., 2014](#) [\(e.g., Clements et al., 1992](#) [Rusakov and Kullmann, 1998](#) [Budisantoso et al., 2013](#) [Matthews et al., 2022](#) [Mendonca et al., 2022](#)).

Maximum glutamate peak concentrations recorded during the cue period were significantly higher in GTs than in STs (phenotype: $F(1,28.85) = 8.85$, $P = 0.006$, $\eta_p^2 = 0.23$; **Fig. 5c**). In contrast, maximum peak amplitudes locked to other task events all were significantly higher in STs (missed turns: $F(1,193) = 55.20$, $P < 0.001$, $\eta_p^2 = 0.22$, **Fig. 5d**; cued stops: $F(1,27.67) = 33.32$, $P < 0.001$, $\eta_p^2 = 0.55$, **Fig. 5e**; reward delivery (during cued turn trials): $F(1,19.28) = 28.88$, $P < 0.001$, $\eta_p^2 = 0.40$, **Fig. 5f**; note that errors in response to stop cues – false turns – were extremely rare and thus not analyzed). In addition, in GTs, turn cue-locked glutamate concentrations rose faster (time to first peak: phenotype: $F(1,19.38) = 7.10$, $P = 0.02$, $\eta_p^2 = 0.27$, GTs: means: 0.7 s; STs: 0.8 s; not shown) and peaked less frequently than in STs (number of peaks: $F(1,27.81) = 13.80$, $P < 0.001$, $\eta^2 = 0.33$, means: GTs: 1.3 peaks; STs: 1.6; not shown). Significant phenotype-specific effects on time-to-peak and the number of peaks locked to other task events were less consistent (missed turns: faster peak times in GTs ($F(1,193) = 5.43$, $P = 0.02$) but similar peak numbers; cued stops: similar peak times but more peaks in GTs ($F(1,29.21) = 18.24$, $P < 0.001$); reward delivery: similar peak times and numbers; not shown).

Glutamate trace characteristics predicting cued turns

The characteristics of glutamate traces (maximum peak concentration, number of peaks, and time to peak) were extracted from 548 recordings of turn cue trials, 364 of which yielded a turn (GTs: 206, STs: 158) and 184 a miss (GTs: 112, STs: 72), to determine whether such characteristics, individually or in combination, disproportionally predicted more cued-triggered turns in GTs or STs. Contingency tables were used to compare phenotype and outcome-specific proportions and to compute the probability for turns in GTs relative to STs.

Turn cue-evoked glutamate peak concentrations greater than 2.8 μM were followed by disproportionally more turns in GTs than STs (2.8–8 μM : $P = 0.045$ – < 0.0001). With increasing glutamate maximum threshold concentrations, GTs were 1.12 ($> 2.8 \mu\text{M}$) to 1.40 ($> 8 \mu\text{M}$) times as likely as STs to turn (circles in **Fig. 6a**; concentrations $> 10 \mu\text{M}$ yielded contingency table cell counts of $n < 10$ and thus were not included).

Furthermore, if one single glutamate peak occurred following the onset of turn cues and during the 2-s cue presentation period, GTs were 1.43 times as likely as STs to turn ($P < 0.0001$). In addition, if only one single glutamate peak followed the turn cue, increasing amplitudes of these peaks predicted significantly higher relative probabilities for turns in GTs, plateauing at $> 4 \mu\text{M}$ where GTs were 1.73 times as likely as STs to turn (rhombi in **Fig. 6a**; note that all data points of this curve reflect significant phenotype-dependent differences in the proportion of turns and misses).

In contrast to the significantly higher odds for turns in GTs following a single cue-evoked glutamate peak, following 2 and 3 glutamate peaks (no more than 3 turn cue-locked peaks were observed), GTs were only 0.74 times as likely as STs to turn ($P < 0.0001$). However, in the presence of 2 or 3 turn cue-locked glutamate peaks (peaks occurring during the cue presentation period), increasing maximum peak levels again increased the probabilities of GTs to turn, so that at maximum peak threshold levels $> 4 \mu\text{M}$, GTs were as likely as STs to turn (both $P > 0.85$; triangles in **Fig. 6a**). Representative traces of 4 selected data points (see data points labeled b,c,d,e in **Fig. 6a** are shown in **Figs. 6b–e**).

Irrespective of whether a single or multiple glutamate peaks were evoked by the turn cue, increasing maximum glutamate concentrations increased the relative probabilities of GTs to turn. This observation was substantiated by the finding that the slopes of the linear regressions of both curves (top and lower in **Fig. 6**) were different from zero (both $P < 0.0001$). However, the absolute differences between single-versus multi-peak-contingent probabilities are noteworthy. Comparing

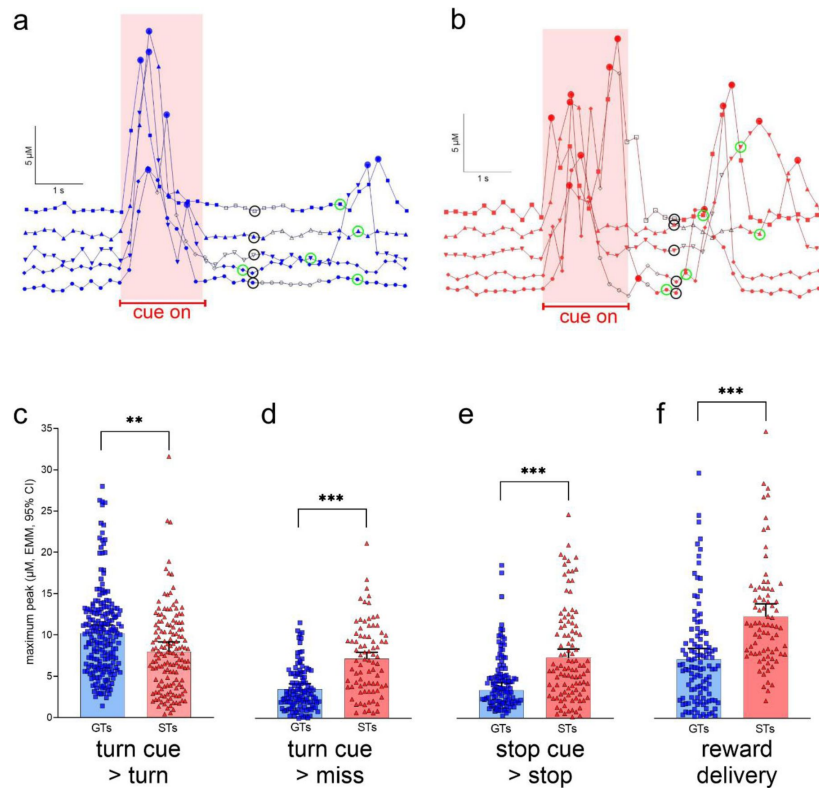


Figure 5

Glutamate concentrations locked to turn and stop cues and reward delivery. **a** and **b** show representative recording traces from GTs and STs (5 each), respectively, recorded during cued turns. Filled circles indicate peaks (defined in Methods), open black symbols were associated with the execution of the turn, symbols surrounded by a black circle with treadmill stops and the onset of a 5-s pause (see also **Fig. 2d**), and symbols surrounded by a green circle mark the delivery of reward following a cued turn. Currents recorded during the cue-on period (reddish background) were used to determine cue-locked peak characteristics, and those recorded during a 2-s period following reward delivery were used to compute reward-locked glutamate peaks. These current traces illustrate the predominance of single turn cue-locked peaks in GTs (**a**), contrasting with the more frequent presence of 2 or 3 turn cue-locked peaks in STs (**b**). Furthermore, reward delivery more reliably evoked glutamate peaks in STs (**b**). LMM-based analysis of traces collected from N=30 rats (15 females) indicated significantly higher turn cue-locked maximum peaks in GTs when followed by actual turns (**c**). In contrast, maximum glutamate concentrations were higher in STs during all other response categories (missed turns, **d**; cued stops **e**; note that errors following stop cues, that is, false turns, were extremely rare and thus peaks obtained from these trials were not analyzed), and following reward delivery during cued turn trials (**f**; **c-f** depict estimated marginal means (EMMs), and 95% CI; main effects of phenotype: **, ***: $P < 0.01$, 0.001).

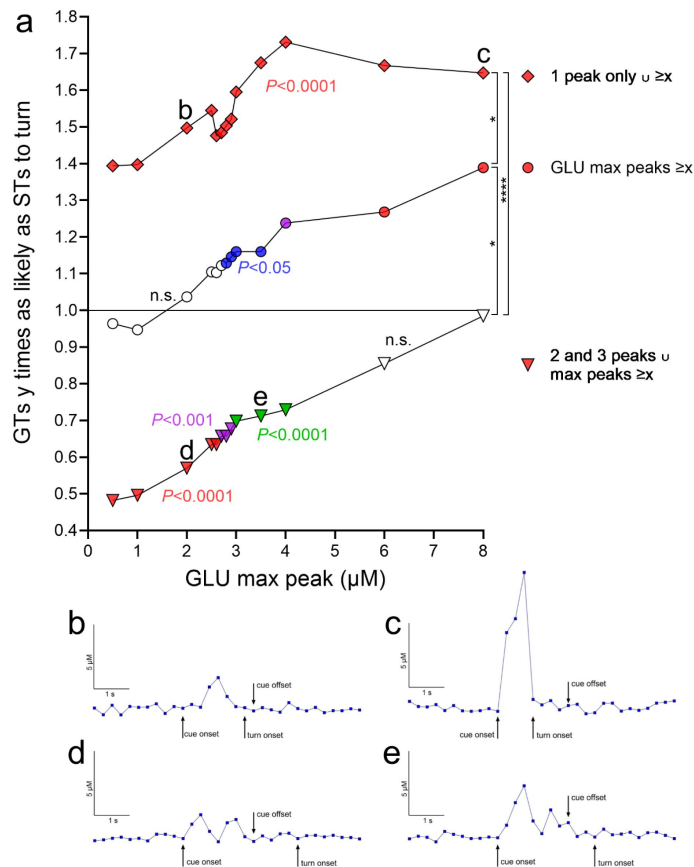


Figure 6

Relative probabilities for cued turns in GTs given the presence of individual and combined properties of turn cue-locked glutamate peaks (based on a total of 548 traces, 364 recorded during cued turns and 184 during misses, 318 from GTs and 230 from STs). **a**: The ordinate depicts relative turn probabilities derived from contingency table analyses. A value of 1 indicates that GTs were as likely as STs to turn (inserted horizontal line), while a value of 2 that GTs were twice as likely as STs to turn. For each relative turn probability value, the associated significance level, derived from contingency table analyses and reflecting the degree of dissimilarity of the proportion of turns and misses in GTs versus STs, is indicated by the symbol color (no fill, not significant, n.s.; blue, $P < 0.05$; green, $P < 0.01$; magenta, $P < 0.001$; red: $P < 0.0001$). The three probability curves reflect predictions based solely on turn cue-locked maximum peak glutamate concentrations (abscissa; middle curve, circles), such concentrations derived from the presence of a single turn cue-locked peak (top curve, rhombi), or such concentrations in conjunction with the presence of multiple (2 or 3) turn cue-locked peaks (bottom curve, triangles). In **a**, the data points labeled b,c,d and e mark data points for which representative traces are shown in **b-e** (all traces are from cued turns in GTs, as the plot in **a** indicates the probability of GTs to turn relatively to STs). For example, the data point next to the label b in **a** indicates that in the presence of a single turn cue-locked glutamate peak of about 2.0 μM , GTs were 1.5 times as likely as STs to turn (significantly different from as likely to turn as STs at $P < 0.0001$). A trace exemplifying this data point is shown in **b**. Circles in **a**: Regardless of other glutamate trace properties, increasing maximum peak concentrations, beginning with 2.8 μM glutamate, yielded significantly different proportions of turns and misses in GTs and STs and rising relative probabilities for GTs to execute a turn ("GLU max peaks $\geq x$ "; the slope of the linear regression of all three curves was significantly different from zero; see Results). For example, for maximum peak concentrations $\geq 4 \mu\text{M}$, GTs were 1.24 times as likely as STs to turn. Rhombi in **a**: The presence of a single turn cue-locked peak strongly increased the relative probability for GTs to turn (see Results). In conjunction with increasing amplitudes of these peaks ("1 peak only $\cup \geq x$ "), these probabilities did further increase, reaching, for example, 1.73 for glutamate concentrations $\geq 4 \mu\text{M}$. Triangles in **a**: In striking contrast, the presence of 2 or 3 turn cue-evoked peaks (no more than 3 turn cue-locked peaks were observed) significantly lowered the relative probability for a turn in GTs below 1 or, conversely, indicated that STs were more likely to turn as GTs. However, in conjunction with rising max amplitude threshold levels ("2 and 3 peaks $\cup$ max peaks $\geq x$ "), the relative probabilities of GTs to turn increased, so that at maximum peak concentrations of $\geq 6 \mu\text{M}$ glutamate, the proportions of turns and misses no longer differed significantly between the phenotypes. The vertical brackets on the right symbolize significant differences between the relative probabilities of the three trace characteristics ($H(3)=33.41$, $P < 0.0001$; Kruskal-Wallis test; multiple comparisons: *, ****, $P < 0.05$, 0.0001).

the relative turn probabilities at maximum peak concentrations $>4 \mu\text{M}$, GTs were 1.002 times *more* likely (or nearly exactly twice *as* likely) as STs to turn if the number of cue-evoked glutamate peaks was limited to one (rhombi in [Fig. 6a](#)) when compared to the presence of 2 or 3 peaks (triangles in [Fig. 6a](#)).

Together, these analyses of glutamate trace characteristics indicated that increasing maximum peak glutamate concentrations and the presence of a single, cue-evoked glutamate peak strongly increased the relative probability of GTs to execute a cued turn. Moreover, the combination of these two properties yielded significantly even higher relative turn probabilities in GTs ([Fig. 6](#)). In contrast, the presence of multiple glutamate peaks, GTs were significantly less likely than STs to turn although, when combined with high ($<4 \mu\text{M}$) maximum peak concentrations, turn proportions of the two phenotypes no longer differed significantly. These findings suggest fundamentally different turn cue-evoked glutamate release dynamics in GTs versus STs, perhaps involving separate afferent circuitry influencing the excitability of cortico-striatal glutamatergic terminals. Therefore, we predicted that inhibition of the cortico-striatal neurons affects turning rates and cued turn-evoked glutamate peaks primarily in GTs, consistent with prior evidence indicating their relative reliance on top-down, cortico-fugal systems to execute cued responses.

Inhibition of fronto-striatal projection disrupts cued turning and turn cue-evoked glutamate in GTs, but not STs

As described above, in GTs, cued turning was associated with tightly orchestrated single glutamate release events that were closely locked to the turn cue, comparable with attended cue-evoked cholinergic transients in cortex ([Parikh et al., 2007b](#); [Gritton et al., 2016](#); [Howe et al., 2017](#)), and therefore hypothesized to reflect cortico-striatal activation. In contrast, in STs, more slowly rising, multiple glutamate release events, as well as the presence of relatively greater reward delivery-locked glutamate release, may have reflected the impact of intra-striatal circuitry and ascending, including dopaminergic, inputs on the excitability of glutamatergic terminals of cortico-striatal projections (e.g., [Tritsch and Sabatini, 2012](#); [Agnoli et al., 2013](#); [Dautan et al., 2020](#); [Moss et al., 2021](#)). We tested the hypothesis that turn cue-locked glutamate peaks in GTs reflect cortico-striatal activation by using a dual vector approach to express an inhibitory DREADD in fronto-striatal projection neurons.

Following PCA screening, GTs ($n=12$, 6 females) and STs ($n=10$, 4 females) underwent surgery to infuse an inhibitory DREADD vector into the prelimbic cortex and a Cre-expressing, retrogradely transported plasmid into the DMS (see [Fig. 7](#) for a timeline and a schematic illustration of the dual vector strategy used to inhibit cortico-striatal projections). An additional group of GTs ($n=5$, 2 females) and STs ($n=2$, 2 females) received infusions of a mCherry expressing control vector into the prelimbic cortex, in addition to the Cre-expressing plasmid into the DMS, to allow for the assessment of potential off-target effects of CNO. Following this surgery, all rats were trained to CTTT performance criterion. Thereafter, the effects of CNO or vehicle on CTTT performance were assessed in $n=14$ rats (9 GTs, 3 females, 5 STs, 4 females; [Fig. 7c](#)). The remainder of the rats (8 GTs, 4 females, 7 STs, 2 females) underwent a second surgery to implant glutamate recording electrodes into the DMS ([Fig. 7d](#)), followed by the assessment of the effects of CNO on performance and glutamate transients ([Fig. 7d](#)).

CNO disrupts cued turns in GTs

On the first of 4 days of testing of the effects of CNO or vehicle, and when the vehicle for CNO was administered, the number of turns/turn cue trials and the number of stops/stop cue trials did not differ significantly between animals without ([Fig. 7c](#)) and with ([Fig. 7d](#)) implanted MEAs (both $t < 0.65$, both $P > 0.53$, at $\alpha = 0.05/2$). Thus, for the initial analysis of the effects of CNO on

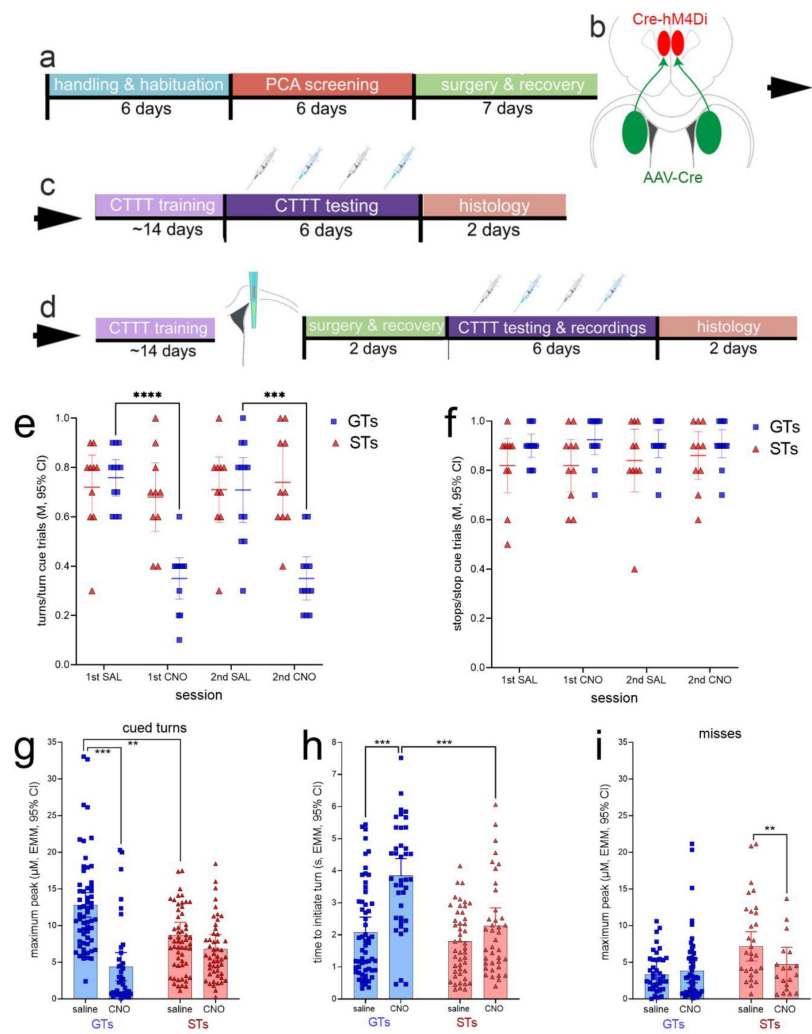


Figure 7

Timeline of experimental procedures and effects of CNO on CTTT performance. **a:** Following initial handling and screening to identify the rat's phenotype (17 GTs, 7 females; 12 STs, 6 females), rats were infused with either a Cre-dependent inhibitory DREADD or the empty control vector (both expressing mCherry; **b**) into the lower layers of the prelimbic cortex, and a retrogradely transported, Cre-expressing plasmid into the DMS (expressing eGFP). These rats continued to undergo CTTT performance training, followed either by (**c**) a test of the effects of vehicle or CNO (the blue syringes symbolizing CNO administration) or by (**d**) implantation of an MEA for the measurement of glutamate concentrations into the mediodorsal stratum. Following recovery from that surgery, the effects of CNO and vehicle on performance and performance-associated glutamate concentrations were assessed. In GTs, administration of CNO significantly reduced the relative number of cued turns (**e**), but not cued stops (**f**; graphs show individual data, mean and 95% CI; *post hoc* multiple comparisons: ***,****: $P < 0.001$, 0.0001, Tukey's Honest Significant Difference test). CNO had no effects on cued turns or stops in rats expressing an empty control vector (not shown; see Results). In GTs, CNO administration attenuated cue-locked maximum peak glutamate concentrations during (residual) cued turns (**g**) and also increased the number of peaks (not shown). Residual turns took significantly more time to be initiated in GTs than STs, as was indicated by an analysis based on turn initiation times from individual trials (**h**). CNO administration did not affect maximum peak glutamate concentrations in GTs during missed turns (**i**), cued stops, false turns and locked to reward delivery (not shown; EMMs, and 95% CI; multiple comparisons: *, **, ***: $P < 0.05$, 0.01, 0.001).

CTTT performance, data from both subgroups were combined. The effects of CNO/vehicle in GTs and STs on cued turns and cued stops were analyzed individually, using ANOVA of the effects of phenotype and day (repeated measures ANOVA, with $\alpha=0.05/2$).

In rats expressing the inhibitory DREADD, administration of CNO reduced the relative number of cued turns in GTs but not STs (main effect of phenotype: $F(1,20)=9.45$, $P=0.006$; main effect of day: $F(3,60)=17.27$, $P<0.0001$; phenotype $\times$ day: $F(3,60)=16.30$, $P<0.0001$; see [Fig. 7e](#) for multiple comparisons). As reflected by the size of the effect of CNO in GTs ($\eta^2=0.72$), the number of cued turns in GTs were approximately halved by the administration of CNO. In contrast, CNO did not affect the relative number of cued stops (main effect of day: $F(3,60)=0.18$, $P=0.89$; day $\times$ treatment: $F(3,60)=0.26$, $P=0.86$; main effect of phenotype: $F(1,20)=5.14$, $P=0.04$, n.s. at $\alpha=0.05/2$, reflecting a trend for relatively more cued stops in GTs; [Fig. 7f](#)). In rats expressing the empty control vector, CNO affected neither the relative number of cued turns (all $F<2.14$, all $P>0.17$) nor cued stops (all $F<1.13$, all $P>0.43$; not shown).

CNO attenuates turn cue-locked glutamate peaks in GTs

Guided by the results illustrated in [Figs. 5](#) and [6](#) and using identical peak definitions and extraction methods, the analyses of the effects of CNO on glutamate transients focused on cue-locked maximum peak concentrations and the number of peaks. Linear mixed-effects models were used to analyze the effects of CNO or vehicle on glutamate transients recorded from GTs and STs expressing the inhibitory hM4Di DREADD or the empty control construct. Glutamate peaks from cued turns, missed turns, cued stops, and false turns were analyzed ([Fig. 7g-j](#)).

Administration of CNO significantly reduced turn cue-locked maximum peak glutamate concentrations in GTs, but not STs (phenotype $\times$ treatment: ($F(1,200.79)=18.42$, $P<0.001$; main effect of treatment: ($F(1,200.79)=44.76$, $P<0.001$; phenotype: ($F(1,9.50)=0.73$, $P=0.41$). Multiple comparisons (shown in [Fig. 7g](#)) confirmed that CNO significantly suppressed turn cue-locked maximum peaks in GTs ($\eta^2=0.21$), but not STs, and that following the administration of vehicle, GTs again (see [Fig. 5c](#)) exhibited higher cue-locked maximum peak levels than STs. CNO reduced maximum peaks in GTs to statistically similar levels seen in STs irrespective of treatment (mean, SEM: GTs: 8.6 ± 0.7 μM ; STs: 7.8 ± 0.7 μM ; [Fig. 7g](#)).

As seen before, the number of glutamate peaks seen during cued turns was significantly higher in STs than in GTs (main effect of phenotype: $F(1,11.72)=8.10$, $P=0.02$; mean, SEM: GTs: 1.7 ± 0.2 peaks; STs: 2.4 ± 0.2). CNO administration increased the number of peaks only in GTs, as was indicated by a significant interaction between the effects of treatment and phenotype and multiple comparisons ($F(1,73)=3.83$, $P=0.04$, $\eta^2=0.05$; means, SEM: GTs/SAL: 1.5 ± 0.2 peaks; GTs/CNO: 2.0 ± 0.2 ; STs/SAL: 2.3 ± 0.2 ; GTs/CNO: 2.4 ± 0.2 ; not shown).

Residual turns following CNO in GTs

CNO robustly reduced cued turn rates in GTs ([Fig. 7e](#)). Furthermore, we observed CNO-attenuated maximum glutamate peaks ([Fig. 7g](#)) during (residual) cued turns, raising the question whether these residual turns differed behaviorally from those seen in vehicle-treated GTs. Therefore, we compared residual cued turns following CNO with turns executed following vehicle administration and, guided by the turn initiation time data shown in [Fig. 3e](#) (baseline performance), focused on this measure. Based on data from individual trials ([Fig. 7h](#)), administration of CNO significantly increased the time GTs took to initiate turns, relative to cue onset, when compared with STs (LMM; phenotype $\times$ treatment: $F(1,173.72)=9.70$, $P=0.002$, $\eta^2=0.05$; see [Fig. 7h](#) for multiple comparisons).

Effects of CNO on glutamate peaks during misses, cued stops and reward delivery

In contrast to turn-cue locked maximum glutamate concentrations in trials followed by turns, when followed by misses, CNO had no effect (main effect of treatment: $F(1,133.99)=1.62$, $P=0.21$; **Fig. 7i**). However, a significant interaction between the effects of treatment and phenotype ($F(1,133.99)=3.88$, $P=0.04$; $\eta^2=0.03$; **Fig. 7i**) reflected that CNO reduced cue-locked maximum peak concentrations in STs in trials yielding misses (results of multiple comparisons are shown in **Fig. 7i**).

There were no effects of CNO on stop cue-locked glutamate peaks, irrespective of whether the stop cue was followed by a stop or a (false) turn (all $P>0.23$; not shown. As before (**Fig. 5f**), peak glutamate concentrations locked to reward delivery were higher in STs than in GTs (main effect of phenotype: $F(1,10.45)=22.38$, $P<0.001$, $\eta^2=0.68$). CNO administration resulted in a significant decrease of reward-locked glutamate maximum peak concentrations in both phenotypes (main effect of treatment: $F(1,412.73)=4.83$, $P=0.03$, $\eta^2=0.01$; interaction: $F(1,412.729)=1.35$, $P=0.25$; mean, SEM: vehicle: 10.3 ± 0.6 μM ; CNO: 9.1 ± 0.3 μM ; not shown).

Importantly, in control rats expressing the empty construct in cortico-striatal projection neurons, administration of CNO had no effect on any parameter of cue-locked glutamate peaks (main effects of treatment and interactions: all $F<0.96$, all $P>0.33$; not shown).

DREADD expression in prelimbic cortex predicts CNO efficacy

The efficacy of the expression of eGFP in the DMS and of mCherry in the prelimbic cortex was quantified using categories that are illustrated in **Figs. 8a-f**. Furthermore, we computed the proportion of neurons in the prelimbic cortex which co-expressed both fluorochrome reporters (**Fig. 8h,i**). As the area of eGFP expression in neuronal soma in the DMS may only partially correspond to the synaptic space of prelimbic projections, it was not unexpected that DMS eGFP transfection efficacy scores did not correlate significantly with the efficacy of CNO to reduce cued turns (GTs: $R^2=0.01$, $P=0.84$; STs: $R^2=0.32$, $P=0.09$). In contrast, in GTs, but not STs, the expression efficacy of mCherry in the prelimbic cortex, that is, of the inhibitory DREADD construct, was significantly correlated with the efficacy of CNO to attenuate cued turns (GTs: $R^2=0.48$, $P=0.02$; STs: $R^2=0.06$, $P=0.51$; **Fig. 8g**). Likewise, in GTs, but not STs, the proportion of neurons expressing both the reporter for the inhibitory DREADD and the retrogradely transported Cre-expressing construct was positively correlated with CNO-induced reduction of cued turns (GTs: $R^2=0.77$, $P=0.0004$; STs: $R^2=0.06$, $P=0.51$; **Fig. 8j**). Thus, the degree of CNO-induced inhibition of prelimbic cortex-DMS projections predicted the efficacy of CNO to disrupt cued turning solely in GTs.

Discussion

The present experiments assessed the role of cortico-striatal glutamate signaling for cued movement and movement suppression. Furthermore, GTs and STs were selected for the expression of opponent cognitive-motivational biases, to test the hypothesis that cue-locked cortico-striatal glutamate signaling contributes to cued movements preferably in GTs. The impact of individual differences can be illustrated by contrasting the actual results with hypothetical findings obtained had experiments been conducted in non-selected rats. 1) Turn cue-evoked glutamate concentrations reached higher levels in GTs compared with STs. In contrast, glutamate concentrations during missed turns, stops, and locked to reward delivery, reached significantly higher levels in STs. In the absence of phenotype as a factor, cue-evoked glutamate concentrations merely would have been found to be higher during turns when compared with missed turns and stops, but not reward delivery. 2) Turn cue-evoked glutamate peak concentrations, in conjunction

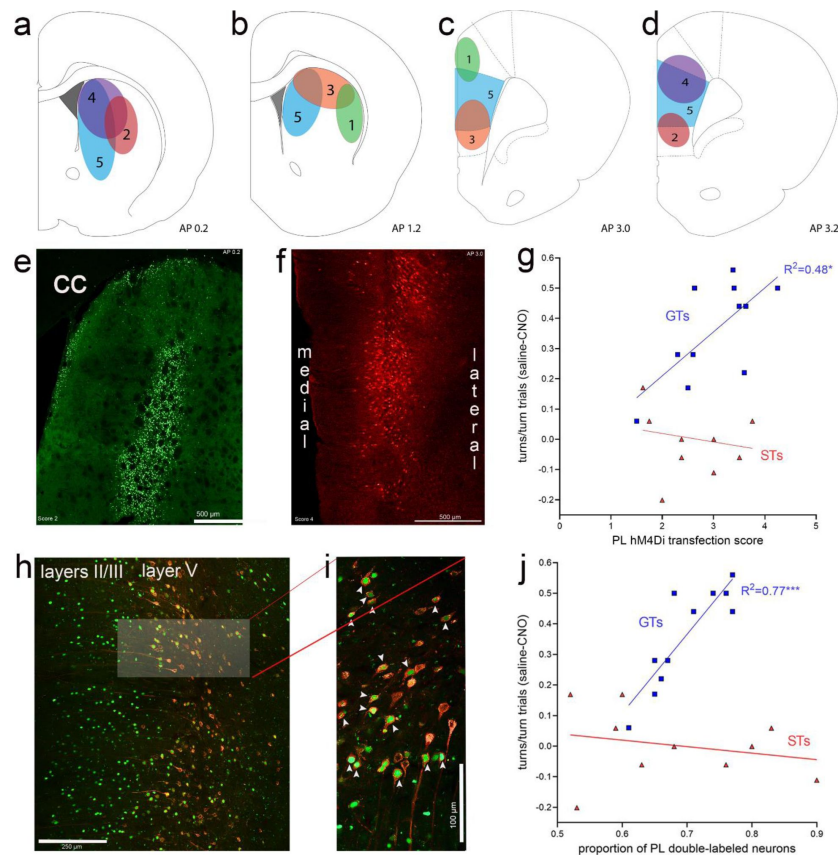


Figure 8

Expression of eGFP (green channel), indicating the expression of the retrograde Cre-expressing vector in the DMS and, following retrograde transport, in the medial frontal cortex. The presence of the amplified mCherry fluorescent reporter signal (red channel) in cortex indicated the expression of the inhibitory hM4Di DREADD vector. **a-d**: Illustration of expression efficacy ratings for eGFP in the DMS (**a,b**) and of mCherry in the prelimbic cortex (prelimbic cortex; **c,d**), with a top score of 5 indicating complete or near-complete expression exclusively in the DMS projection field of prelimbic cortex efferent neurons (Mailly et al., 2013) and prelimbic cortex, respectively. The example of eGFP expression in the DMS in **e** (cc, corpus callosum) received a score of 2 (illustrated in **a**) because the expression field was located in part lateral to the prelimbic cortex projection field. **f** depicts an example of mCherry expression in the prelimbic cortex that was assigned a scores of 4 as it was restricted to the deeper layers in the prelimbic cortex and extended dorsally into cingulate cortex. In GTs, but not STs, mCherry expression scores in cortex were significantly correlated with the efficacy of CNO to reduce cued turns (**g**; note that the ordinate in **g** depicts the difference between the proportion of turns in vehicle-minus CNO-treated rats, so that higher scores indicate greater CNO effects, and therefore higher expression scores were correlated with greater CNO effects). **h** depicts the retrogradely transported eGFP and the expression of mCherry, including the co-expressing of both fluorochromes, primarily in layer 5 of the prelimbic cortex. The brightened rectangular region in **h** is enlarged in **i** and shows neurons co-expressing nucleolar eGFP and cytoplasmic mCherry (white arrows in **i**). In GTs, but not STs, the proportion of double-labeled neurons in prelimbic cortex was significantly correlated with the efficacy of CNO to attenuate cued turns (**j**; note that higher scores on the ordinate depicts greater CNO effects).

with single spikes, nearly doubled the likelihood for a turn in GTs relative to STs. In contrast, in conjunction with multiple glutamate spikes, relatively low glutamate peak concentrations predicted that STs were more likely to turn. In the absence of phenotype as a factor, some animals would have been found to turn when peak glutamate concentrations were relatively low but multiple peaks were present, and some when the concentrations of single spikes reached the same relatively low peak concentrations. Consequently, the number of glutamate spikes would have been concluded to have no impact on turn probability. 3) In GTs, but not STs, inhibition of cortico-striatal neurons reduced cued turn rates and turn cue-evoked glutamate peak concentrations, and it increased the number of glutamate peaks. In the absence of phenotype as a factor, the effects of CNO would have been found to be extremely variable, and the collective results would not have provided an interpretational framework for such a finding. 4) In GTs, but not STs, the efficacy of CNO to attenuate cued turns was significantly correlated with transcription efficacy scores for the inhibitory DREADD, and with the relative number of prelimbic-DMS neurons expressing the inhibitory DREADD. In the absence of phenotype as a factor, no such relationships would have been found (both $R^2 < 0.17$), limiting the attribution of limited behavioral and glutamatergic effects of DREADD activation to prelimbic cortex-DMS projections. These contrasts between actual and hypothetical results indicate the essential role of the phenotypes for the discussion of the present evidence.

Goal- and sign-tracking are behavioral indices that predict the presence of larger, opponent cognitive-motivational biases (for review see [Sarter and Phillips, 2018](#)). STs preferably assign motivational significance to reward cues, with the result that they perceive such cues as rewarding, worth approaching and working for. This bias by STs is illustrated by their propensity to approach and contact Pavlovian food or drug cues (e.g., [Robinson and Flagel, 2009](#)), and by the relatively greater efficacy of cocaine cues to instigate drug taking (e.g., [Saunders and Robinson, 2011](#); [Yager and Robinson, 2013](#)). In contrast, GTs are biased toward an analysis of the behavioral utility of a reward or drug cue. While, in GTs, such cues likewise evoke an enhanced motivational state and the expectation of reward, GTs do not approach and contact such cues. Such a top-down bias indexed by goal-tracking can be revealed not only by the absence of non-instrumental behavior toward reward cues. In contrast to STs, GTs were shown to be able to utilize complex occasion setters predicting the availability of cocaine ([Saunders et al., 2014](#); [Pitchers et al., 2017a](#)). Research in humans has confirmed the presence of such broad opponent cognitive-motivational styles indexed by sign- and goal-tracking, particularly the propensity of STs to attend to cues preferably as a function of cue salience and prediction of reward ([Schad et al., 2020](#); [Duckworth et al., 2022](#); [Colaizzi et al., 2023](#)).

The present findings indicate that turn cue processing in the striatum of GTs is closely controlled by cortico-striatal activity, and that GTs utilize turn cues more reliably than STs. These findings complement prior evidence from experiments assessing the selection of cues for extended processing and behavioral control. GTs out-performed STs in tasks taxing the attentional detection of cues, in part by deploying goal-directed attentional mechanisms to detect cues and maintain stable task performance (e.g., [Paolone et al., 2013](#); [Kucinski et al., 2018](#); [Phillips and Sarter, 2020](#); [Kucinski et al., 2022](#)). The present results are consistent with a relatively less efficacious cue detection process in STs that may be mediated primarily via deficient cortical cholinergic mechanisms ([Paolone et al., 2013](#); [Koshy Cherian et al., 2017](#); [Pitchers et al., 2017b](#); [Kucinski et al., 2022](#); [Carmon et al., 2023](#)). Consequently, subcortical mechanisms predominately contribute to behavioral control in STs (see also [Flagel et al., 2011](#); [Campus et al., 2019](#); [Schad et al., 2020](#); [Iglesias et al., 2023](#)).

In GTs, inhibition of prelimbic-DMS projections attenuated cued turns and associated glutamate concentrations. In contrast, suppression of movement, as seen during cued stops, and associated glutamate release dynamics, remained unaffected by CNO and, therefore, may not depend on cortico-striatal processing of movement cues (see also [Vandaele et al., 2019](#); [Cruz et al., 2022](#)). In GTs, turn cue-evoked glutamate concentrations as well as the presence of single peaks may be

speculated to be secondary to the cue detection-mediating generation of single cholinergic transients in prelimbic cortex and the resulting generation of high-frequency oscillations (Gritton et al., 2016 [DOI](#); Howe et al., 2017 [DOI](#)). In the DSM, single high-amplitude glutamate transients from cortical afferents may recruit multiple populations of interneurons and striatal output neurons, thereby forming, for a precise period of time, fronto-striatal cell assemblies to force cue-associated action selection (see also Hart et al., 2018a [DOI](#); Hart et al., 2018b [DOI](#); Oberto et al., 2022 [DOI](#)).

Cortico-striatal glutamatergic terminals are also recruited, directly and indirectly, via striatal interneurons and striatal dopaminergic afferents, thereby supporting bidirectional glutamatergic-dopaminergic interactions and allowing reward expectation and action outcome to influence the glutamatergic representation of movement cues (Bamford et al., 2004 [DOI](#); Agnoli et al., 2013 [DOI](#); Kosillo et al., 2016 [DOI](#); Cai and Ford, 2018 [DOI](#); Moss et al., 2021 [DOI](#); Choi et al., 2023 [DOI](#); Holly et al., 2024 [DOI](#)). Such striatal control of cortico-striatal terminals, primarily via cholinergic and dopaminergic heteroreceptors (for review see Lovinger et al., 2022 [DOI](#)), may preferentially yield suppression of action in STs. This view is consistent with several findings and considerations. 1) Glutamate concentrations in STs were relatively higher during stops, irrespective of the validity of stops (false stops or stop cue-evoked stops). 2) The presence of multiple cue-locked glutamate peaks in STs (a second or third peak occurred on average 1.58 ± 0.39 s and 1.83 ± 0.25 s (M, SD), respectively, following cue onset), may reflect relatively slow, reverberating interactions between striatal interneurons, dopamine signaling, and cortico-striatal glutamate release (e.g., Dorst et al., 2020 [DOI](#); Frost Nylén et al., 2021). In GTs, the rare presence of multiple peaks may have interfered with cortico-striatal cue import and therefore reduced the efficacy of cued turning. 3) Inhibition of the prelimbic-DSM pathway impacted neither turn cue-locked glutamate release in STs nor cued stops or stop cue-locked glutamate release in either phenotype. 4) Reward delivery-locked peak glutamate concentrations were higher in STs than GTs. Thus, in STs, these glutamatergic transients may be speculated to primarily signal the expectation of reward, and also the heightened perceptual value of a reward predicting cue (see also Pitchers et al., 2017b [DOI](#); Saunders et al., 2018 [DOI](#); Iglesias et al., 2023 [DOI](#); Bernklau et al., 2024 [DOI](#)).

Given that CNO did not attenuate turn cue-evoked glutamate concentrations in STs, the neuronal source for striatal information about the sensory properties of the cue remains unclear. Other striatal afferent systems, originating from midbrain cholinergic and glutamatergic regions (Dautan et al., 2014 [DOI](#); Assous et al., 2019 [DOI](#); Dautan et al., 2020 [DOI](#)), or thalamus (Díaz-Hernández et al., 2018 [DOI](#)), may serve as alternative sources to “import” post-perceptual cue information into striatal circuitry to support well-practiced, habitual cued action in STs and action suppression in both phenotypes.

The present results contribute to bio-behavioral conceptualizations of the impact of individual differences in the vulnerability for compulsive drug taking (e.g., Volkow et al., 2006 [DOI](#); Robbins et al., 2012 [DOI](#); Marhe et al., 2013 [DOI](#); Kilts et al., 2014 [DOI](#); George and Koob, 2017 [DOI](#); Pitchers et al., 2018 [DOI](#)). In contrast, the significance of cognitive-motivational endophenotypic traits such as goal- and sign-tracking for neurological disorders has yet to be explored (but see Deik et al., 2012 [DOI](#); Kucinski et al., 2018 [DOI](#)). Consistent with the view that striatal pathology contributes little to the explanation of individual, clinical severity of PD (Johansson et al., 2024 [DOI](#)), our findings predict a greater vulnerability of PD patients who preferentially utilize cortico-striatal mechanisms to process movement cues, that is, GTs, for cholinergic loss-associated cued movement control and falls (Sarter et al., 2014 [DOI](#); Sarter et al., 2021 [DOI](#)). In contrast, STs may generally be less efficient in executing cued movements, particularly in situations featuring complex and competing movement cues (e.g., Beaulne-Seguin and Nantel, 2016 [DOI](#)). The investigation of the impact of endophenotypic indices that predict individual differences in fundamental brain function may be useful to reveal potentially predictive neuronal and cognitive markers of psychiatric and neurological disease vulnerability.

Conflict of Interest

Authors report no conflict of interest.

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Reviewer #1 (Public review):

Summary:

The authors measured glutamate transients in the DMS of rats as they performed an action selection task. They identified diverse patterns of behavior and glutamate dynamics depending on the pre-existing behavioral phenotype of the rat (sign tracker or goal tracker). Using pathway-specific DREADDs, they showed that these behavioral phenotypes and their corresponding glutamate transients were differentially dependent on input from the prefrontal cortex to the DMS.

Strengths:

Overall there are some very interesting results that make an important contribution to the field. Notably, the results seem to point to differential recruitment of the PL-DMS pathway in goal-tracking vs sign-tracking behaviors.

Weaknesses:

There is a lot of missing information and data that should be reported/presented to allow a complete understanding of the findings and what was done. The writing of the manuscript was mostly quite clear, however, there are some specific leaps in logic that require more elaboration, and the focus at the start and end on cholinergic neurons and Parkinson's disease are, at the moment, confusing and require more justification.

<https://doi.org/10.7554/eLife.100988.1.sa3>

Reviewer #2 (Public review):**Summary:**

The authors aimed to determine whether goal-directed and cue-driven attentional strategies (goal- and sign-tracking phenotypes) were associated with variation in cued motor responses and dorsomedial striatal (DMS) glutamate transmission. They used a treadmill task in which cues indicated whether rats should turn or stop to receive a reward. They collected and analyzed several behavioral measures related to task performance with a focus on turns (performance, latency, duration) for which there are more measures than for stops. First, they established that goal-trackers perform better than sign-trackers in post-criterion turn performance (cued turns completed) and turn initiation. They used glutamate sensors to measure glutamate transmission in DMS. They performed analyses on glutamate traces that suggest phasic glutamate DMS dynamics to cues were primarily associated with successful turn performance and were more characteristic of goal-trackers (ie. rats with "goal-directed" attentional strategy). Smaller and more frequent DMS glutamate peaks were associated with other task events, cued misses (missed turns), cued stops, and reward delivery and were more characteristic of sign-trackers (i.e. rats with "cue-driven" attentional strategies). Consistent with the reported glutamate findings, chemogenetic inhibition of prelimbic-DMS glutamate transmission had an effect on goal-trackers' turn performance without affecting sign-trackers' performance in the treadmill task.

Strengths:

The power of the sign- and goal-tracking model to account for neurobiological and behavioral variability is critically important to the field's understanding of the heterogeneity of the brain in health and disease. The approach and methodology are sound in their contribution to this important effort.

The authors establish behavioral differences, measure a neurobiological correlate of relevance, and then manipulate that correlate in a broader circuitry and show a causal role in behavior that is consistent with neurobiological measurements and phenotypic differences.

Sophisticated analyses provide a compelling description of the authors' observations.

Weaknesses:

It is challenging to assess what is considered the "n" in each analysis (trial, session, rat, trace (averaged across a session or single trial)). Representative glutamate traces (n = 5 traces (out of hundreds of recorded traces)) are used to illustrate a central finding, while more conventional trial-averaged population activity traces are not presented or analyzed. The latter would provide much-needed support for the reported findings and conclusions. Digging deeper into the methods, results, and figure legends, provides some answers to the reader, but much can be done to clarify what each data point represents and, in particular, how each rat contributes to a reported finding (ie. single trial-averaged trace per session for multiple sessions, or dozens of single traces across multiple sessions).

Representative traces should in theory be consistent with population averages within phenotype, and if not, discussion of such inconsistencies would enrich the conclusions drawn from the study. In particular, population traces of the phasic cue response in GT may resemble the representative peak examples, while smaller irregular peaks of ST may be missed in a population average (averaged prolonged elevation) and could serve as a rationale for more sophisticated analyses of peak probability presented subsequently.

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Reviewer #3 (Public review):

Summary:

Avila and colleagues investigate the role of glutamate signaling in the dorsomedial striatum in a treadmill-based task where rats learn to turn or stop their walking based on learning cue-associations that allow them to acquire rewards. Phenotypic variation in Pavlovian conditioned sign and goal-tracking behavior was examined, where behavioral differences in stopping and turning were observed. Glutamate signals in the DMS were recorded during the treadmill task and were related to features of cue-controlled movement, with a stronger relationship seen for goal trackers. Finally, chemogenic inhibition of prelimbic neurons projecting to the DMS (the predicted source of those glutamate signals), preferentially affected cued movement in goal trackers. The authors couch these experiments in the context of cognitive control-attentional mechanisms, movement disorders, and individual differences in cue reactivity.

Strengths:

Overall these studies are interesting and are of general relevance to a number of research questions in neurology and psychiatry. The assessment of the intersection of individual differences in cue-related learning strategies with movement-related questions - in this case, cued turning behavior - is an interesting and understudied question. The link between this work and growing notions of corticostriatal control of action selection makes it timely.

Weaknesses:

The clarity of the manuscript could be improved in several places, including in the graphical visualization of data. It is sometimes difficult to interpret the glutamate results, as presented, in the context of specific behavior, for example.

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Author response:**Reviewer #1 (Public Review):***Strengths:*

Overall there are some very interesting results that make an important contribution to the field. Notably, the results seem to point to differential recruitment of the PL-DMS pathway in goal-tracking vs sign-tracking behaviors.

Thank you.

Weaknesses:

There is a lot of missing information and data that should be reported/presented to allow a complete understanding of the findings and what was done. The writing of the manuscript was mostly quite clear, however, there are some specific leaps in logic that require more elaboration, and the focus at the start and end on cholinergic neurons and Parkinson's disease are, at the moment, confusing and require more justification.

In the revised paper, we provide additional information in support of results and clarify procedures and findings. Furthermore, we expand the discussion of the proposed interpretational framework that suggests that the contrasts between the cortical-striatal

processing of movement cues in sign- versus goal trackers are related to previously established, parallel contrasts in the cortical cholinergic detection of attention-demanding cues.

Reviewer #2 (Public review):

Strengths:

The power of the sign- and goal-tracking model to account for neurobiological and behavioral variability is critically important to the field's understanding of the heterogeneity of the brain in health and disease. The approach and methodology are sound in their contribution to this important effort.

The authors establish behavioral differences, measure a neurobiological correlate of relevance, and then manipulate that correlate in a broader circuitry and show a causal role in behavior that is consistent with neurobiological measurements and phenotypic differences.

Sophisticated analyses provide a compelling description of the authors' observations.

Thank you.

Weaknesses:

It is challenging to assess what is considered the "n" in each analysis (trial, session, rat, trace (averaged across a session or single trial)). Representative glutamate traces (n = 5 traces (out of hundreds of recorded traces)) are used to illustrate a central finding, while more conventional trial-averaged population activity traces are not presented or analyzed. The latter would provide much-needed support for the reported findings and conclusions. Digging deeper into the methods, results, and figure legends, provides some answers to the reader, but much can be done to clarify what each data point represents and, in particular, how each rat contributes to a reported finding (ie. single trial-averaged trace per session for multiple sessions, or dozens of single traces across multiple sessions).

Representative traces should in theory be consistent with population averages within phenotype, and if not, discussion of such inconsistencies would enrich the conclusions drawn from the study. In particular, population traces of the phasic cue response in GT may resemble the representative peak examples, while smaller irregular peaks of ST may be missed in a population average (averaged prolonged elevation) and could serve as a rationale for more sophisticated analyses of peak probability presented subsequently.

Figures 5c-f depict individual data from all rats and trials. For all major analyses, the revised manuscript consolidates information about the number of rats per phenotype and sex, and the number of trials contributed by individual rats, in the result section.

As detailed in the section on statistical methods, and as mentioned by the reviewer under Strengths, we used advanced statistical methods to assure that data from individual animals contribute equally to the overall result, and to minimize the possibility that an inordinate number of trials obtained from just one or a couple of rats biased the overall analysis.

As the reviewer correctly pointed out, we have chosen not to show trial- or subject-averaged traces to illustrate glutamate release dynamics across trials. The present analyses focus on peak glutamate concentrations, the number of peaks, and the timing of peaks relative to a task cue or a behavioral event. Within a response bin, such as the 2-s period following turn cues, glutamate peaks – as defined in Methods - occur at variable times relative to cue onset. Averaging traces over a population of rats or trials would “wash-out” the phenotype- and task

event-dependent patterns of glutamate peaks, yielding, for example, a single, nearly 2-s long plateau for cue-locked glutamate recordings from STs (Figure 5b). Thus, subject- or trial-averaged traces would not illustrate the major findings described in this paper and would rather be uninformative. As already mentioned, individual data from all subjects and trials are shown in Figs 5c-f.

Reviewer #3 (Public review):

Strengths:

Overall these studies are interesting and are of general relevance to a number of research questions in neurology and psychiatry. The assessment of the intersection of individual differences in cue-related learning strategies with movement-related questions - in this case, cued turning behavior - is an interesting and understudied question. The link between this work and growing notions of corticostriatal control of action selection makes it timely.

Thank you.

Weaknesses:

The clarity of the manuscript could be improved in several places, including in the graphical visualization of data. It is sometimes difficult to interpret the glutamate results, as presented, in the context of specific behavior, for example.

We appreciate the reviewer's concerns about the complexity of some of the graphics, particularly the results from the arguably innovative analysis illustrated in Figure 6. Figure 6 illustrates that the likelihood of a cued turn can be predicted based on single and combined glutamate peak characteristics. The revised legend for this figure provides additional information and examples to ease the readers' access to this figure.

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